The Atlas of Operational Ideas

Identity, Novelty, and Scientific Discovery in Neuro-Symbolic AI

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Contents

Abstract

Scientific communication is organised around documents, while scientific reasoning is organised around claims, mechanisms, transformations, models, methods, analogies, and evidential relations that cut across documents. This mismatch is becoming more consequential as the literature grows and as artificial intelligence systems are asked not merely to retrieve papers, but to judge whether two contributions contain the same idea, whether a proposal has a genuine precedent, how a method evolved, or where a mechanism might transfer. An atlas of operational ideas would introduce an intermediate epistemic layer between source documents and scientific reasoning. It would represent ideas as task-relative, evidence-grounded structures; distinguish several kinds of identity rather than imposing a universal equivalence relation; record lineages as contestable inferences rather than as automatic consequences of similarity; and treat novelty as a relation to a dated, explicitly bounded state of knowledge.

The project is scientifically feasible in a restricted but important sense. Contemporary language models, scholarly knowledge graphs, provenance standards, computational history of ideas, and recent benchmarks for scientific lineage and novelty supply many of the necessary components. None, however, settles the central conceptual problems. An idea is not simply a phrase, a topic, a proposition, a method name, or a point in an embedding space. The same contribution may be identical to an earlier one with respect to purpose, different with respect to mechanism, analogous with respect to causal organisation, and independent with respect to historical descent. A system that collapses these relations will generate precisely the errors it is intended to prevent.

This monograph develops a research programme for a neuro-symbolic atlas rather than a product specification. It proposes operational ideas as structured objects whose boundaries depend on the question being asked; a typed and query-relative theory of identity; an account of priority as inference under incomplete observation; and a multidimensional conception of novelty that remains distinct from usefulness. The neuro-symbolic architecture is understood as an epistemic division of labour. Neural systems retrieve, align, abstract, and propose. Symbolic structures preserve explicit roles, constraints, arguments, and provenance. Probabilistic and temporal models express uncertainty, competing interpretations, and the fact that new evidence can revise earlier conclusions.

A dedicated chapter investigates Vector Symbolic Architectures and Hyperdimensional Computing as a possible representational substrate. Their binding, bundling, approximate similarity, and factorisation operations offer a promising way to combine structured symbolic content with robust distributed memory. The central hypothesis is that one scientifically meaningful component of novelty can be modelled as failure of structured reconstruction from known components under an explicit grammar

of admissible transformations. This hypothesis is promising but hazardous: a reconstruction residual may indicate real novelty, missing literature, a defective codebook, inappropriate granularity, extraction error, or incoherence. VSA/HDC is therefore treated as an experimental instrument whose claims must be audited against symbolic structures and sources, not as a novelty oracle.

The book concludes with a programme of datasets, benchmarks, temporal backtests, ablations, and human studies capable of falsifying the proposal. A successful atlas could strengthen literature synthesis, attribution, peer review, laboratory memory, cross-domain transfer, and the training of scientific AI. Its deeper contribution would be infrastructural: making the transformations of ideas inspectable, comparable, and open to revision without pretending that the history or identity of ideas can be reduced to a single canonical map.

Preface

The phrase *encyclopaedia of ideas* is attractive because it evokes a missing form of intellectual orientation. Researchers can search for papers, authors, keywords, citations, datasets, and increasingly for passages that answer a natural-language question. They still cannot reliably ask which earlier works contain the same mechanism in different terminology, which part of a proposal is inherited and which part changes, whether an alleged first appearance is only the earliest one visible in a narrow corpus, or whether a distant field has already developed an analogous solution. These are not exotic questions. They are ordinary components of literature review, research design, peer evaluation, and historical attribution. They become difficult because the objects over which they range are not stable database records. They are interpretations of content.

The project examined here should therefore not be understood as an attempt to enumerate timeless atoms of thought. Such a programme would inherit the strongest objections made against older histories of "unit-ideas": it would detach ideas from the contexts in which their meaning and force were constituted, presume that continuity is given rather than argued for, and quietly elevate the editor's preferred decomposition into an ontology of the world (Lovejoy 1936; Skinner 1969; Betti and Berg 2014). Nor is the project simply a larger knowledge graph. Existing scholarly graphs provide indispensable identifiers, citation links, provenance, and increasingly machine-actionable representations of claims, but they do not by themselves determine when differently worded contributions instantiate the same operational principle (Jaradeh et al. 2019; Priem, Piwowar, and Orr 2022).

The central move of this book is more modest and, for that reason, more technically demanding. It treats an idea representation as an explicit model constructed for a purpose. The representation must say what distinctions it preserves, what context it suppresses, what evidence supports it, and what transformations count as continuity for the inquiry at hand. An idea object is not discovered in a document in the way a date or a chemical formula might be extracted. It is proposed as a disciplined abstraction from one or more attestations. That proposal can be compared, challenged, refined, or rejected.

This stance changes how success should be judged. A system need not settle the metaphysics of ideas to be scientifically useful. It must instead improve a determinate class of epistemic tasks while exposing the assumptions that make its answers possible. It should retrieve better precedents than document-level search, distinguish mechanistic equivalence from superficial resemblance, reconstruct lineages without confusing similarity with influence, and make its uncertainty legible. It should become less confident, not more confident, when corpus coverage is poor or when alternative decompositions remain plausible.

The emphasis on scientific depth also determines what has been left out. This is not an application programming interface design, a software roadmap, or an administrative plan. Those matters are downstream of the conceptual research. The difficult questions concern ontology, semantics, compositionality, historical inference, open-world novelty, and the interaction between distributed and symbolic representations. Building a polished service before addressing them would automate ambiguity rather than resolve it.

The chapters proceed from the inadequacy of documents as units of reasoning to operational ideas, typed identity, lineage, novelty, and priority. The neuro-symbolic and VSA/HDC analyses translate those commitments into testable representational hypotheses; the final chapters examine evaluation and impact. The guiding metaphor is a microscope, not an oracle.

1 Documents, Ideas, and the Missing Epistemic Layer

1.1 The document is not the unit of scientific reasoning

Modern scholarly infrastructure is extraordinarily good at treating the document as an object. A paper has an identifier, a title, authors, a venue, references, dates, licences, and often links to data or software. Citation indexes and open scholarly graphs connect millions of such objects. Full-text search and language-model retrieval can locate passages that mention a concept or answer a question. These achievements are foundational. Yet the document remains a container whose boundaries are determined by publication practice rather than by the structure of reasoning.

A single paper usually contains several epistemically different contributions. It may formulate a problem, introduce a mechanism, adapt an existing method, report an implementation, derive a theorem, offer an explanation, define an evaluation protocol, and delimit the conditions under which the claims hold. Conversely, one scientific idea may be distributed across a sequence of papers, correspondence, software releases, experimental protocols, and later reconstructions. The identity of the publication and the identity of the idea therefore cross-cut one another. A paper is neither a minimal unit of contribution nor a maximal unit of intellectual continuity.

This mismatch is partly hidden when the task is retrieval. A useful paper can be found even when the system cannot say exactly which component is relevant. It becomes visible when the task requires comparison. Suppose two methods optimise the same objective but use different mechanisms; two theories invoke the same causal organisation under incompatible vocabularies; or a recent paper presents an old principle in a new domain with the first convincing evidence. Whether these count as the same idea is not answered by textual overlap, topical proximity, or citation links alone. The comparison must preserve some relations and ignore others.

The problem is intensified by the use of artificial intelligence in research. Language models can summarise documents and generate plausible descriptions of novelty, but fluent explanation is not evidence that the underlying judgment is correct. Recent benchmarks make this limitation concrete. RINoBench contains 1,381 research ideas with human novelty judgments and finds that model-generated rationales may resemble expert explanations even while the final judgments diverge substantially from the human reference (Schopf and Färber 2026). IdeaGene-Bench represents papers using typed, evidence-grounded idea objects and lineage differences; across its tasks, the strongest evaluated system reached only 27.3 per cent exact accuracy (Zhou et al. 2026). These results should not be read merely as evidence that present models need more training. They reveal that document understanding, compositional comparison, and historical reasoning are distinct problems.

Scientific memory is also asymmetric. Successful, recent, highly cited, English-language documents are easier to retrieve than negative results, technical reports, older books, local journals, or work preserved in other languages. A system that searches only what is readily indexed will mistake visibility for priority and popularity for conceptual centrality. The resulting error is not simply missing data. It affects the ontology inferred from the available record: familiar ideas appear more coherent, more original, and more influential because the traces that would complicate them are absent.

The desired intermediate layer can be described by contrasting several objects that are often treated as interchangeable. Table 1 does not propose rigid essences. It clarifies which distinctions a system must keep available if it is to answer scientific questions rather than merely organise text.

Table 1. Distinctions that must remain visible in an idea-centred representation.

Epistemic object	Why it cannot be reduced to "an idea" without qualification
Word or phrase	A linguistic expression can be ambiguous, can change meaning through time, and can name several distinct mechanisms.
Topic or concept label	A topic groups discourse, but does not specify a claim, causal organisation, procedure, or contribution.
Proposition or claim	A truth-apt statement captures what is asserted, but may omit the mechanism, purpose, evidence, and method by which it matters.
Mechanism	A mechanism describes organised entities and activities producing a phenomenon, but may support many claims and implementations.
Method or procedure	A method specifies how something is done, yet the same operational principle may admit many procedures and one procedure may combine several principles.
Model or theory	A model structures explanation and prediction at a larger scale than a minimal transferable idea.
Implementation	A concrete realisation carries engineering choices that may be irrelevant to functional or mechanistic identity.

Epistemic object	Why it cannot be reduced to “an idea” without qualification
Operational idea	A task-relative abstraction that records selected roles, relations, conditions, consequences, evidence, and provenance for a stated inquiry.

The table explains why the proposed layer cannot consist of names attached to papers. The left column lists objects that scholarly systems already represent with varying success. The right column identifies the information lost when each object is treated as a complete idea. An operational idea is placed last not because it is metaphysically more fundamental, but because it is an explicit synthesis designed to coordinate the others. Its legitimacy depends on whether the synthesis is useful and revisable.

Figure 1 shows the resulting architecture at the level of epistemic commitments. Documentary sources remain primary. Passages, equations, figures, code fragments, and artefacts become addressable attestations. From them, a system or curator proposes task-relative idea objects. Typed relations connect those objects into families and possible lineages. Only then are higher-level inquiries such as identity, novelty, priority, analogy, and explanation posed. The dashed return path matters as much as the forward path: an inquiry may reveal that the evidence is insufficient and force the system back to the sources.

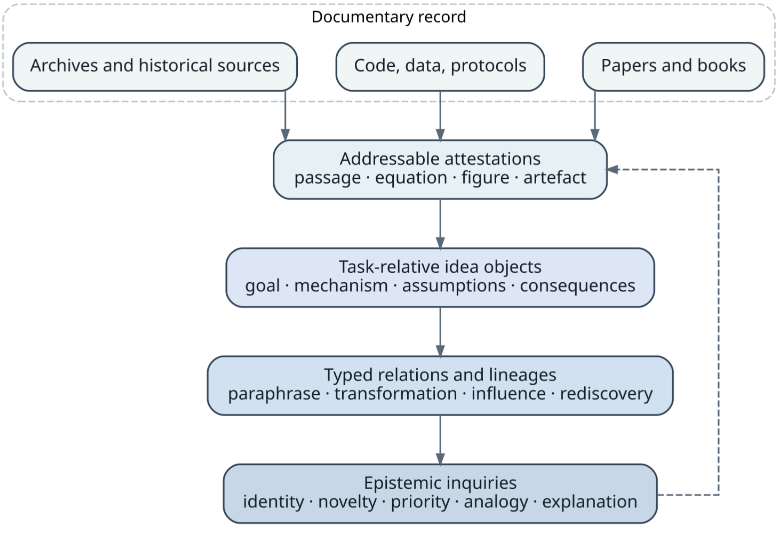

Figure 1. The missing epistemic layer between documentary sources and inquiries about ideas. The return path indicates that comparison can expose missing or inadequate evidence.

This layered picture prevents two common category errors. The first is to treat generated abstractions as if they were present verbatim in the source. The second is to let a similarity score silently stand in for a historical or logical relation. Every movement upward in the figure adds interpretation. Every interpretation must therefore remain linked to the evidence and assumptions that support it.

1.2 Precursors solve important fragments, not the whole problem

The ambition to reorganise knowledge around ideas rather than documents has a long history. Wells imagined a continually revised "world brain," while Bush's memex emphasised associative trails through a growing record (Wells 1938; Bush 1945). Adler's *Syntopicon* indexed major ideas across a selected canon (Adler 1952). These projects recognised that alphabetical catalogues and isolated volumes do not match the associative and comparative structure of inquiry. Their limitation was not lack of imagination. They depended on editorial synthesis that could not be executed or contested at the scale and granularity required for contemporary science.

The history of ideas offers a more direct precedent and a warning. Lovejoy's "unit-ideas" aimed to trace recurrent components across intellectual history (Lovejoy 1936). Contextualist criticism, most prominently associated with Skinner, objected that apparent recurrence can be manufactured by removing statements from the linguistic and practical situations in which they performed different actions (Skinner 1969). Betti and van den Berg later argued that computational history need not choose between unstructured particularism and naive universal units. It can make its interpretive models explicit and study continuities relative to those models (Betti and Berg 2014, 2016). This is close to the position developed here: continuity is not read directly from words but argued through a representation whose commitments are inspectable.

Linguistics and cognitive science contribute several forms of semantic decomposition. Frame semantics represents situations through roles and relations, and FrameNet links these structures to attested language (Fillmore 1982; Baker, Fillmore, and Lowe 1998). Prototype theory and family resemblance challenge the assumption that categories are defined by necessary and sufficient features (Rosch 1975; Wittgenstein 1953). Conceptual spaces model concepts geometrically, while formal concept analysis derives lattices from relations between objects and attributes (Gärdenfors 2000; Ganter and Wille 1999). These traditions show that semantic structure can be made computational without assuming that every category has a single crisp boundary. They do not, by themselves, provide a theory of scientific contribution, lineage, or priority.

Pattern languages are particularly relevant to the phrase "fundamental ideas about how things are done." Alexander and colleagues described recurrent problem–solution structures that could be reused without being copied literally (Alexander et al. 1977). A pattern records context, a recurring tension, a core resolution, and relations to other patterns. This is already more operational than a topic ontology. Yet a pattern language normally abstracts away from the historical evidence needed to determine priority or influence, and it is not designed to compare subtle modifications of mechanisms across scientific literatures.

Knowledge graphs and semantic publishing solve another indispensable fragment. Nanopublications and micropublications make small assertions, provenance, evidence, and argument relations machine-actionable (Groth, Gibson, and Velterop 2010; Kuhn et al. 2018; Clark, Ciccarese, and Goble 2014). The Open Research Knowledge Graph seeks to represent contributions in structured form rather than as undifferentiated prose (Jaradeh et al. 2019). PROV-O supplies a vocabulary for derivation and responsibility, while SKOS supports concept schemes and mappings (World Wide Web Consortium 2013, 2009). These standards make it possible to say who asserted what, from which source, and through which process. They still require a domain theory that determines what the assertion is *about* and when two structured contributions should be aligned.

Scientometrics addresses novelty and influence at a scale inaccessible to close reading. Studies of atypical combinations suggest that high-impact work often mixes a conventional knowledge base with a limited amount of unusual recombination (Uzzi et al. 2013). Other research shows that scientists tend to favour established strategies and that bibliometric indicators can penalise novelty or conflate it with interdisciplinarity (Foster, Rzhetsky, and Evans 2015; Wang, Veugelers, and Stephan 2017). Such work is valuable for population-level patterns. Its proxies operate mainly over citations, journals, keywords, or co-occurrences, and therefore remain several steps removed from mechanistic or functional novelty.

Recent systems move closer to the proposed atlas. Scideator represents papers through facets such as purpose, mechanism, and evaluation and supports recombination grounded in the literature (Radensky et al. 2024). CHIMERA mines instances in which scientific work combines elements from prior concepts and mechanisms (Sternlicht and Hope 2026). Graph2Idea constructs graph-structured contexts of problems, methods, mechanisms, and findings for idea generation (Li, Tu, and Han 2026). IdeaGene introduces minimal typed objects and explicit lineage differences (Zhou et al. 2026). Literature-grounded novelty assessment retrieves and reranks precedents before producing a judgment (Shahid et al. 2025). Together these systems demonstrate that idea-level structure is no longer only philosophical speculation. They also expose the unresolved questions of identity, granularity, evidential support, and evaluation.

Table 2 maps the principal traditions by pairing their strongest contribution with the unresolved issue that prevents them from constituting an atlas on their own. The purpose of the table is not to rank fields. It identifies interfaces between research programmes that have often developed separately.

Table 2. Existing traditions provide complementary components of an idea atlas while leaving different conceptual gaps.

Research tradition	Contribution and remaining gap
Encyclopaedic and associative knowledge systems	They organise material across documents, but rely on editorial categories that are difficult to formalise, scale, and contest.
History of ideas and conceptual history	They supply contextual interpretation and accounts of semantic change, but have limited computational machinery for evidence-grounded comparison at scale.
Frames, prototypes, and conceptual spaces	They model roles, graded similarity, and semantic organisation, but do not directly represent scientific lineage, priority, or evidential force.
Pattern languages and procedural knowledge	They capture reusable problem–solution structures, but usually omit historical provenance and fine-grained transformation relations.
Scholarly knowledge graphs and semantic publishing	They provide identifiers, provenance, claims, and argument links, but leave the cross-document identity of operational ideas under-specified.
Scientometrics and science of science	They measure population-level novelty and influence, but rely on proxies that rarely expose mechanisms or role-wise differences.
LLM-based scientific ideation and review	They retrieve, abstract, and recombine flexibly, but their judgments can be uncalibrated and their explanations need not track correct decisions.
Lineage and recombination benchmarks	They make inheritance and transformation computationally explicit, but remain domain-limited and do not yet settle historical or semantic identity.

The table suggests that the project is integrative rather than wholly unprecedented. Its possible originality lies in binding these components under a common epistemic discipline. That discipline requires evidence-addressed abstractions, typed identity, explicit historical uncertainty, and tests that distinguish genuine improvement from the production of plausible narratives.

1.3 A research claim rather than an encyclopaedic promise

The strongest defensible claim is not that all ideas can be placed into one final taxonomy. It is that many scientific tasks can be improved by representing selected contributions as multi-resolution, evidence-grounded structures and by reasoning explicitly over their transformations. This claim can be decomposed into empirical hypotheses without reducing the project to an engineering checklist.

The first hypothesis concerns retrieval. A system that knows the roles played by components of an idea should find precedents missed by lexical and document-level search. A mechanism may have migrated between fields under different terminology; a recent method may change only an assumption that ordinary similarity treats as peripheral; a concept may have undergone semantic drift that makes historical vocabulary misleading. Structured retrieval should improve not only the relevance of results but the explanation of why a result is a precedent.

The second hypothesis concerns comparison. Document embeddings compress many dimensions of similarity into a single geometry. An idea atlas should produce a typed comparison in which purpose, mechanism, assumptions, evidence, implementation, and lineage can vary independently. The advantage is not merely interpretability. Typed comparison changes the answer. Two contributions may be functionally equivalent but mechanistically distinct, or mechanistically similar but historically independent. A single similarity score cannot preserve these alternatives.

The third hypothesis concerns scientific memory. Citation graphs record declared references, not all intellectual dependence, and they mix many reasons for citation. An evidence-grounded lineage model should reconstruct candidate inheritance relations while distinguishing documentary influence, structural resemblance, and independent rediscovery. It should also retain disagreement. Historical scholarship frequently depends on sources outside the canonical article trail; the system must allow a newly found archive or older-language source to revise the map.

The fourth hypothesis concerns novelty. What is new can be understood partly as what cannot be reconstructed from earlier structures under a stated repertoire of transformations. This conception is not intended to replace expert judgment. It creates an object of analysis: which roles are inherited, which are transformed, which appear unsupported by precedent, and how sensitive that conclusion is to corpus expansion or a change of representation. A good novelty analysis would be less categorical than current rhetoric, not more.

The fifth hypothesis concerns artificial intelligence. Scientific AI requires training signals for distinctions that ordinary next-token prediction does not explicitly encode: paraphrase versus mechanistic equivalence, analogy versus influence, inherited component versus novel insertion, and absence of evidence versus evidence of absence. A curated atlas could provide contrastive examples, provenance-preserving memory, and benchmarks for compositional reasoning. Whether this improves discovery must be demonstrated against strong document-based systems rather than assumed.

These hypotheses imply demanding criteria of success. The system must outperform strong baselines on tasks where structure should matter. It must remain stable enough under corpus growth to be useful, while revising itself when genuinely earlier evidence appears. It must report the kinds of uncertainty that affect an answer rather than compressing them into generic confidence. Experts must be able to inspect the relation between a conclusion and its sources. Most importantly, the representation must fail informatively. When it cannot distinguish true novelty from missing evidence or model mismatch, that ambiguity should appear in the output.

Three levels of claim must remain separate. The project does not need the ontological thesis that ideas possess unique natural boundaries. It does require a representational thesis: for specified questions, some decompositions preserve scientifically relevant invariants better than alternatives. It also advances an epistemic thesis: making those decompositions, alignments, and sources explicit can improve inquiry. A stronger institutional thesis - that such representations should guide review, funding, or credit - would require additional evidence about bias, contestability, and effects on behaviour. Conflating these levels would allow success on a narrow benchmark to masquerade as a theory of thought or a mandate for scientific governance.

Feasibility will also vary by domain. Fields with explicit formalisms, executable procedures, stable artefacts, and well-dated corpora offer favourable conditions for early experiments. Mechanisms can be compared against equations, code, protocols, or causal diagrams rather than inferred from prose alone. Domains in which meaning depends heavily on rhetorical situation, tacit practice, or contested interpretation are harder, but they are not therefore irrelevant. They provide stress tests for whether the atlas can preserve context and disagreement instead of forcing false precision. A credible programme should begin where claims are testable and then examine how far its abstractions travel.

The burden of proof is consequently comparative rather than metaphysical. The atlas must show that its intermediate objects enable distinctions, retrievals, or revisions that are difficult to obtain from documents, embeddings, and citation graphs alone. It must also show that the gain is not purchased by hiding curatorial labour or importing expert conclusions into the benchmark. The research contribution lies in a disciplined account of what is represented, how it is supported, when it changes, and where it ceases to be reliable.

This standard separates a scientific instrument from an encyclopaedic promise. An encyclopaedia tends to present a settled view of its entries. An atlas, in the intended sense, presents a navigable and revisable model of a terrain that exceeds any one representation. Different maps can preserve different invariants. The existence of several legitimate maps is not a defect if their purposes and transformations are explicit. It becomes a defect only when the system silently treats one map as the terrain itself.

2 What Kind of Thing Is an Idea?

2.1 Ontological pluralism and operational ideas

Any attempt to build an atlas of ideas encounters an immediate circularity. The system needs units in order to compare, trace, and index ideas, but the identity of those units is itself part of what the system is supposed to discover. If the project begins with a fixed inventory, it risks encoding the designers' intuitions as facts. If it refuses all prior structure, it cannot decide what to extract or how to evaluate a match. The appropriate response is not to eliminate interpretation but to represent it explicitly.

The word *idea* covers several ontological categories. In ordinary scientific speech it may refer to a proposition, a conceptual distinction, a mathematical construction, a causal mechanism, an experimental design, an algorithmic strategy, an explanatory schema, or a conjectured possibility. These objects have different identity conditions. Propositions are often compared through logical content. Mechanisms are compared through organised entities and activities that produce a phenomenon (Machamer, Darden, and Craver 2000; Craver and Darden 2013). Procedures are compared through sequences, dependencies, and control conditions. Models may be compared through structure-preserving mappings, predictive consequences, or explanatory roles. An atlas that forces all of them into one undifferentiated type will be simple only at the cost of being wrong.

Ontological pluralism does not imply that comparison is impossible. It means that the comparison must begin by asking what kind of object is under consideration and what relation is sought. The same historical episode can support several representations. The development of backpropagation, for example, can be narrated as a mathematical technique for efficient differentiation, a learning mechanism, an algorithmic procedure, an engineering practice, or a lineage of independent rediscoveries. Each representation selects different evidence and different boundaries. A robust atlas should allow them to coexist and should record mappings between them rather than declaring one to be the uniquely correct idea.

A second source of difficulty is semantic holism. The significance of a scientific statement often depends on a network of assumptions, practices, and contrasts. An expression may retain its words while changing its inferential role. Conversely, two formulations may use entirely different vocabularies while occupying comparable positions in their respective theories. Contextualist history is right to insist that extracting a sentence from its setting can change what act the author performed with it (Skinner 1969). Computational representation must therefore distinguish an attested expression from an interpreted content and from the role that content plays in a broader system.

Prototype and family-resemblance theories offer a useful caution against classical definitions (Rosch 1975; Wittgenstein 1953). Many idea families may have no feature shared by every member. Their continuity can consist of overlapping similarities, recurrent transformations, and a lineage of adaptations. This does not make the family arbitrary. It changes the structure of its identity from a fixed checklist to a network of constrained relations. The identity of "evolutionary computation," "Bayesian inference," or "attention" may be organised around prototypes and historical pathways rather than necessary and sufficient properties.

The project must also separate cognitive ontology from representational utility. Human beings may or may not store ideas as discrete symbolic structures; the atlas does not require a claim about the ultimate format of thought. It requires external representations that support reliable inquiry. A map of a mechanism can be useful even if the brain does not encode mechanisms in the same way. This distinction is particularly important for VSA/HDC. High-dimensional vectors may provide an effective computational medium without thereby revealing the metaphysical nature of ideas.

An operational ontology should therefore be layered. At the documentary level, the system records what sources contain. At the interpretive level, it proposes structures that make comparison possible. At the historiographical level, it records claims about continuity, influence, and priority. At the pragmatic level, it represents what distinctions matter for a particular inquiry. These layers constrain one another but should not be collapsed. A located passage is not yet an idea object; an idea object is not yet a lineage; a lineage is not yet an attribution of credit.

This pluralist position has a practical consequence for data design. The atlas should permit more than one decomposition of the same source and should treat disagreement between decompositions as information. Two curators may agree on the cited passage and disagree on whether it contains one mechanism or two. A model may propose a coarse abstraction useful for cross-domain analogy while a domain expert prefers a fine decomposition needed for experimental replication. Preserving both views, together with the question they serve, is more scientifically faithful than forcing premature consensus.

The term *operational idea* names a representational commitment, not a newly discovered natural kind. It is an abstraction designed to be small enough that its components can be compared and transformed, yet complete enough to preserve an intelligible contribution. Its structure should capture at least the roles that make a difference to the intended inquiry. A convenient starting form is

$$I_Q = \langle G, M, C, A, O, E, P \rangle_Q.$$

Here, G represents a goal, problem, or explanatory target; M a mechanism, transformation, or governing principle; C the context or domain; A relevant assumptions and preconditions; O expected outcomes, consequences, or capabilities; E the evidential support; and P provenance. The subscript Q is essential. It indicates that the boundary and salience of the object depend on a question or family of tasks. The tuple is not a universal definition of every idea. It is a schema for making the choices of representation explicit.

Consider a method that uses random projections to preserve approximate distances. A historical inquiry may represent the mathematical guarantee, its earliest formulation, and later rediscoveries. An engineering inquiry may foreground the input regime, computational cost, distortion bound, and implementation. A cross-domain analogy system may abstract further, retaining only the principle that a high-dimensional object can be mapped to a lower-dimensional space while approximately preserving relational structure. These are not merely different descriptions of a fixed object. They may establish different units because different components can vary independently under the relevant comparison.

The goal role prevents a mechanism from being detached from the problem it addresses. The same operation can count as different ideas when it serves different purposes, and different operations can instantiate the same functional idea. The mechanism role prevents functional equivalence from erasing causal or procedural differences. Context and assumptions prevent a representation from claiming generality that the source does not support. Outcomes connect an idea to what it enables or predicts. Evidence distinguishes a speculative possibility from a validated result. Provenance preserves the path from abstraction back to the record and the process by which the abstraction was produced.

The tuple should not be understood as seven text fields. Each role may itself be a structured object. A mechanism can contain entities, activities, ordering, feedback, and boundary conditions. An assumption can be logical, statistical, material, or institutional. Evidence can include experiments, proofs, simulations, datasets, negative results, and replications. Provenance must identify not only a source but the location and version from which a claim was derived, the agent or model that proposed the interpretation, and any subsequent challenges. The apparent simplicity of the notation is therefore a discipline of organisation, not a denial of complexity.

Operational representation also requires negative and contrastive information. An idea is often understood by what it changes relative to a predecessor. A paper may preserve the goal and evaluation while removing an assumption, replacing a mechanism, or extending the admissible context. If the atlas stores only positive summaries, it will lose the transformation that constitutes the contribution. Similarly, a failed application can reveal a boundary condition more clearly than a successful one. The object should therefore be capable of recording exclusions, limitations, counterexamples, and unresolved alternatives.

This structure differs from a summary. A summary compresses a document for reading; an operational idea supports comparison and inference. It must use stable roles and typed relations that can be aligned across sources. It also differs from an ontology entry. An ontology normally locates a concept within a controlled vocabulary; an operational idea records an organised contribution, its evidence, and its relation to other contributions. Finally, it differs from a claim graph. A claim may be one component of an idea, but the idea can also contain a problem framing, a method of intervention, and a pattern of consequences that no single proposition captures.

The requirement of task relativity prevents the schema from becoming another rigid universalism. The same source can generate several I_Q objects, each with a declared resolution and purpose. The system should retain relations between these representations, such as “refines,” “abstracts,” “focuses on evidence,” or “focuses on mechanism.” This creates a lattice or graph of views rather than a single canonical entry. Formal concept analysis and conceptual-space approaches suggest ways to organise such perspectives, but the atlas must add evidence and transformation history (Ganter and Wille 1999; Gärdenfors 2000).

Task relativity does not mean that anything goes. A representation is constrained by source fidelity, predictive or explanatory usefulness, intersubjective agreement, and performance on downstream tasks. A decomposition that invents an unsupported mechanism is invalid even if it is convenient. A representation so coarse that it cannot distinguish known counterexamples is inadequate for the task. A representation so fine that every wording becomes unique defeats the purpose of comparison. The scientific problem is to discover useful regimes between these extremes.

2.2 Granularity, compositionality, and the boundaries of an idea

The question “What is the smallest idea?” is misleading because minimality is relative to admissible transformations. A component is operationally atomic when it can vary independently in the comparisons of interest and when further decomposition no longer improves those comparisons enough to justify the added complexity. Atomicity is therefore a property of a model and a task, not an intrinsic indivisibility of thought.

This definition resembles the use of modules in engineering and mechanisms in philosophy of science. A mechanism can be decomposed into organised parts until the explanatory question no longer requires finer detail. The sodium channel may be a component in one explanation and a complex mechanism in another. The same is true of scientific ideas. “Gradient-based learning” may be one component in a broad history of optimisation, while a technical analysis separates differentiation, update rule, stochastic sampling, regularisation, and stopping criteria.

Granularity creates a bias in novelty judgments. At a coarse resolution, many contributions appear to repeat familiar ideas. At a fine resolution, almost every implementation appears novel. Neither conclusion is reliable without a theory of which differences are scientifically consequential. The atlas should therefore compare candidates across several resolutions and report where the novelty claim emerges. A contribution that is old at the level of mechanism but new at the level of evidence or capability should not be forced into a binary label.

Compositionality introduces a related problem. Some ideas can be represented as combinations of reusable components, but the combination may create properties not predictable from the parts in isolation. Conceptual blending and research on analogy emphasise that mappings and interactions can generate new structure (Fauconnier and Turner 2002; Gentner 1983). A system that decomposes too aggressively may lose the relational organisation that constitutes the idea. Conversely, a system that treats every whole as irreducible cannot recognise inheritance or recombination.

The appropriate unit is therefore often a structured configuration rather than a bag of components. Two methods may contain the same elements but organise them differently. Feedback, ordering, nesting, and conditional dependence can be decisive. For this reason, the transformation grammar of the atlas must include not only insertion and deletion but substitution, reordering, abstraction, specialisation, change of role, transfer of context, and alteration of constraints. A new idea may consist less in adding a component than in changing the relation between familiar ones.

Context poses another boundary problem. Too little context produces false equivalence; too much context prevents transfer. A mechanism that works in a controlled laboratory setting may fail in an open social system because agency, adaptation, and measurement change the causal structure. Yet retaining every local detail would make cross-domain analogy impossible. The atlas needs representations in which context can be progressively abstracted while the consequences of that abstraction are recorded. One view may state that a mechanism is valid under a narrow set of assumptions; another may propose a functional analogy at a higher level and mark the untested correspondence.

Table 3 summarises characteristic granularity failures. Each row pairs a representational choice with the epistemic distortion it creates. The table is not a catalogue of implementation bugs. It describes failure modes that should become experimental variables in the research programme.

Table 3. Granularity choices create systematic forms of false identity and false novelty.

Representational choice	Characteristic distortion
Treating the whole document as one idea	Independent contributions, inherited components, and local modifications become inseparable.
Treating every sentence or claim as an idea	Relational organisation and procedural structure disappear, while wording differences are mistaken for conceptual difference.
Using one fixed level of abstraction	The system alternates between false equivalence in some tasks and false novelty in others.
Decomposing only into named components	Novel organisation, ordering, feedback, and constraint changes are missed.

Representational choice	Characteristic distortion
Retaining all contextual detail	Cross-document alignment and cross-domain transfer collapse because no two cases are sufficiently similar.
Removing context too early	Analogies become superficial and mechanisms are transferred beyond their valid conditions.
Enforcing a single curator-approved decomposition	Disagreement and alternative explanatory perspectives are erased rather than studied.
Allowing unrestricted decompositions	Comparisons become non-reproducible because any desired relation can be manufactured after the fact.

The table shows why multi-resolution representation is not an optional convenience. It is the only coherent response to the fact that scientific contributions have nested structure and that different inquiries preserve different invariants. Multi-resolution does, however, increase the burden of evaluation. The system must learn when a coarse alignment is informative and when it hides a decisive difference. It must also prevent users from selecting a resolution merely because it supports a desired novelty claim.

One promising principle is *contrastive adequacy*. A representation is adequate for a question when it distinguishes the cases experts regard as importantly different and groups cases they regard as equivalent for that purpose. This can be tested using carefully constructed pairs that vary one dimension at a time. Another principle is *transformational economy*: prefer a representation in which observed lineages can be described by a relatively small and stable set of meaningful transformations, without forcing unrelated cases into the same family. These principles turn granularity into an empirical research problem.

The resulting conception is deliberately anti-essentialist but not anti-formal. An idea object is a model with declared scope, evidence, and resolution. It can be encoded, compared, and evaluated. Its boundaries are open to revision, but revision is constrained by source fidelity and performance. The atlas would therefore map not only ideas but the alternative ways in which ideas can be individuated. This reflexive layer is necessary because any system capable of judging novelty must also be capable of questioning the units on which that judgment depends.

3 Identity Without a Universal *sameAs*

3.1 Identity profiles and typed relations

The question “Are these the same idea?” invites a binary answer that is rarely adequate. Scientific identity is indexed to a purpose. A historian may ask whether two authors formulated the same proposition, an engineer whether two methods realise the same function, a philosopher whether they posit the same mechanism, and a reviewer whether a new submission makes a contribution beyond a cited baseline. The answer can change without contradiction because the relation being tested has changed.

A useful representation begins with an identity profile rather than a universal equivalence relation. For two operational idea objects x and y, the system can construct

$$\Sigma_Q(x, y) = \langle s_{surface}, s_{prop}, s_{func}, s_{mech}, s_{context}, s_{impl}, s_{evid}, s_{lineage} \rangle_Q.$$

Each component describes alignment along a distinct dimension: linguistic surface, propositional content, function or goal, mechanism, context and assumptions, implementation, evidential status, and historical lineage. The values need not all be scalar similarities. Some are structured correspondences, some are categorical relations, and some are probability distributions over competing interpretations. The subscript again records that the comparison is made for a question Q.

This profile separates cases that a generic embedding would place near one another. Two texts may be paraphrases but rely on different evidence. Two algorithms may be functionally equivalent while using distinct computational mechanisms. Two experiments may instantiate the same design but answer different questions. Two mechanisms may be structurally analogous while their components have no historical relation. The profile also prevents a common historiographical error: treating similarity in present-day reconstruction as proof that later authors inherited an idea from earlier ones.

The relevant notion of identity can be expressed through invariants. A comparison question selects a set of roles and relations that must be preserved, together with a set of transformations that are allowed to vary. Informally, x and y count as the same for Q when the selected invariants are preserved under an admissible alignment. A compact notation is

$$x \equiv_Q y \quad \mathit{only\ if} \quad Inv_Q(x) \cong Inv_Q(y) \ \mathit{under\ an\ admissible\ transformation\ in}\ T_Q.$$

The symbol $\cong$ is deliberately weaker than literal equality. It represents a structure-preserving correspondence whose definition depends on the object type. The set T_Q may include renaming, change of notation, abstraction, specialisation, substitution of an implementation, or transfer to a new context. It should not be assumed to form a mathematical group. Scientific transformations are often partial, irreversible, and dependent on preconditions. Treating them as a general transformation grammar is safer than imposing algebraic properties they do not possess.

Identity judgments must also be asymmetric in some cases. "Generalises," "implements," "is derived from," and "repairs a limitation of" are directional. Even apparent equivalence may be asymmetric when one representation contains strictly more detail. A coarse functional description can be satisfied by several fine mechanisms, but the fine mechanisms are not thereby equivalent to one another. The atlas therefore needs both equivalence-like families and directional transformation relations.

Figure 2 illustrates the logic. Two idea objects are not compared in the abstract. They enter through a comparison question that determines which invariants matter. Their purposes, mechanisms, assumptions, implementations, and evidence are aligned separately. The output is a typed judgment that records what is preserved and what changes. The figure is intentionally not a funnel toward a single number. It depicts identity as an interpretive composition of role-wise results.

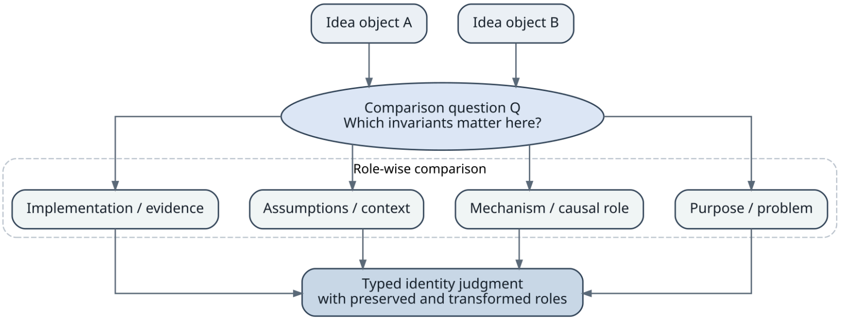

Figure 2. Identity is produced by a question-sensitive comparison of roles, not by a universal similarity threshold.

This account supports both crisp and graded relations. Logical equivalence under a formal language may admit a proof. Functional or mechanistic equivalence often remains graded because descriptions are incomplete and contexts differ. Gradedness should not be confused with vagueness alone. It can represent uncertainty about evidence, partial preservation of structure, or genuine degrees of match. The system should identify which of these interpretations applies.

An important research question is whether identity profiles can be learned without collapsing into latent similarity. Supervision must therefore include contrastive cases in which texts are similar but mechanisms differ, and cases in which texts are dissimilar but a domain expert recognises the same causal organisation. Evaluation should test the correctness of the alignment itself, not merely agreement with a final label. A model that predicts "same mechanism" for the wrong reasons will fail when transferred to new contexts.

A universal sameAs relation is attractive because it simplifies graphs. Once two nodes are declared identical, their properties can be merged. In an atlas of ideas this operation would be dangerous. The identity of an idea depends on what information is inherited across the relation, and different relations license different inferences. A paraphrase permits linguistic substitution. Functional equivalence supports expectations about outputs under stated conditions. Mechanistic equivalence supports a stronger inference about how those outputs are produced. Historical influence supports claims about lineage and credit. None can automatically substitute for the others.

Typed relations also make disagreement more tractable. Experts may agree that two methods solve the same problem and disagree about whether their mechanisms are equivalent. Instead of forcing a single decision, the atlas can store the shared functional relation and competing mechanistic interpretations. A later experiment or more precise reconstruction may resolve the dispute. This is closer to scientific practice than a database constraint that demands immediate identity or separation.

Table 4 sets out a core relation vocabulary and, in the second column, the evidential burden associated with each relation. The purpose is to show that relation types are not merely labels. They determine what sources and arguments are required before the relation should be accepted.

Table 4. Typed identity and lineage relations carry different inferential permissions and evidential requirements.

Typed relation	Meaning and evidential burden
Paraphrase or terminological reformulation	The expressions communicate substantially the same content; linguistic and contextual analysis is required, but historical dependence is not implied.
Propositional equivalence or entailment	The claims have the same truth conditions, or one entails the other, within an explicit formalisation and background theory.
Functional equivalence	The objects pursue the same goal or produce comparable outcomes under stated conditions, even if their mechanisms differ.
Mechanistic equivalence	The organised causal or computational roles correspond at the selected level of abstraction; evidence must support the mapping of entities, activities, and dependencies.
Implementation equivalence	Concrete realisations match in operationally relevant choices; this is stronger than sharing a named method and weaker than literal code identity.
Analogy	A relational structure maps between domains while object-level identities differ; structural fidelity and the limits of transfer must be stated.

Typed relation	Meaning and evidential burden
Specialisation or generalisation	One object narrows or broadens the domain, assumptions, or consequences of another; the direction of inclusion must be justified.
Composition or recombination	A later object integrates components from several antecedents; the contribution may lie in the relations among components rather than in the components alone.
Historical influence	Documentary or contextual evidence supports a causal path of intellectual transmission; semantic similarity is insufficient.
Independent rediscovery	Strong structural correspondence exists while available evidence supports separate development or fails to establish transmission.
Contradiction or replacement	A later object rejects, invalidates, or substitutes for a component of an earlier one; the scope of the conflict must be localised.

The table prevents a semantic graph from becoming overconfident through transitivity. Paraphrase and formal equivalence may be transitive under controlled conditions. Analogy is not generally transitive. Historical influence is directional and can pass through intermediaries, but absence of a direct citation does not settle whether influence occurred. Functional equivalence may fail when contexts are combined. Each relation needs its own composition rules and exceptions.

Typed relations also support a more precise concept of idea families. A family need not be an equivalence class in which every member is mutually identical. It can be a connected subgraph organised around prototypes, recurrent mechanisms, transformations, and lineage. One branch may preserve a mechanism while changing its purpose; another may preserve the purpose while replacing the mechanism. The family is intelligible through the pattern of transformations, not through a feature present in every node.

This conception resembles evolutionary metaphors, including the “idea genomes” proposed in recent lineage work (Zhou et al. 2026), but the metaphor must be used carefully. Biological inheritance has material mechanisms and relatively well-defined units of reproduction. Intellectual inheritance can be indirect, collective, reconstructed retrospectively, and mediated by languages and institutions. Ideas do not reproduce independently of agents and practices. The value of the metaphor lies in explicit comparison of inherited, modified, lost, imported, and newly inserted components, not in claiming a literal genetics of thought.

A typed graph also enables reasoning about counterfactual identity. Researchers may ask whether a method would remain “the same” if a particular assumption were removed, if its implementation were changed, or if it were transferred to another domain. Such questions reveal the implicit invariants of a concept. They can be formalised as controlled transformations of an idea object and evaluated through expert judgments. This creates a bridge between conceptual analysis and machine learning: the model is trained not only on existing pairs, but on interventions that probe the boundary of the category.

3.2 Similarity, lineage, influence, and independent rediscovery

Similarity is evidence for comparison, not a conclusion about history. This distinction is easy to state and difficult to preserve computationally. Retrieval systems naturally rank earlier texts by semantic proximity. When a highly similar predecessor is found, a generated explanation can slide from "resembles" to "derives from," especially if the later work cites related literature. An atlas must keep three questions separate: whether structures are similar, whether one work was historically available to the other, and whether there is evidence of actual transmission.

Historical lineage is an inference from heterogeneous evidence. Direct citation is relevant but neither necessary nor sufficient. Authors cite work for background, criticism, convention, strategic positioning, or reviewer requests. They may inherit an idea through textbooks, colleagues, code, lectures, or a shared research programme without citing the original source. Conversely, two groups may arrive independently at closely related structures because the problem strongly constrains the solution space. The atlas should therefore treat influence as a hypothesis supported by a bundle of temporal, documentary, social, and semantic evidence.

Priority disputes demonstrate why this matters. The earliest published formulation, the earliest private conception, the first public presentation, the first proof, and the first effective implementation may belong to different people. Scientific credit is also shaped by visibility and institutional power. Merton's analysis of priority and the Matthew effect shows that recognition is a sociological process as well as a chronological one (Merton 1957, 1968). Stigler's law of eponymy captures the recurring fact that discoveries are often named after someone other than the earliest contributor (Stigler 1980). A computational atlas should illuminate these distinctions rather than produce a single "inventor" field.

Independent rediscovery deserves a first-class relation because it has scientific value. Repeated emergence can indicate that an idea is a natural solution under shared constraints, that multiple traditions possessed compatible components, or that the time was ripe because enabling evidence or technology had appeared. Collapsing rediscoveries into one node would erase information about the structure of discovery. Treating them as entirely unrelated would miss their conceptual convergence. The atlas should connect them through typed structural identity while keeping their historical paths distinct.

Lineage reconstruction can be represented as an alignment between predecessor and successor objects. The alignment identifies preserved roles, transformations, external imports, and unresolved components. Recent work such as IdeaGene-Bench makes this “diff” perspective operational (Zhou et al. 2026). Its scientific importance is broader than benchmark performance. A lineage difference forces the system to specify what exactly changed. It can reveal that an apparently new theory preserves the explanatory mechanism while changing only the evidence, or that a seemingly incremental method removes an assumption that had constrained an entire class of applications.

The lineage graph should be non-monotonic. New evidence can revise an edge. A newly digitised book may become the earliest known attestation; an archive may establish influence that was previously classified as independent rediscovery; a technical reconstruction may show that two methods thought equivalent rely on incompatible mechanisms. Instead of overwriting the prior state, the atlas should preserve versions, the evidence available at each time, and the reasons for revision. This is essential for scientific accountability and for studying how the atlas itself evolves.

Lineage should be analysed through competing historical hypotheses rather than inferred from one similarity score. For a pair of related works, the main alternatives may include direct transmission, transmission through an intermediary, inheritance from a shared tradition, convergence under common problem constraints, and accidental resemblance. Each predicts a different evidential pattern. Direct transmission is strengthened by temporal accessibility, terminological or diagrammatic traces, explicit acknowledgement, archival contact, and a sequence of distinctive shared choices. Convergence is more plausible when the shared structure follows almost inevitably from the problem while peripheral design decisions differ. A common ancestor is favoured when both works reproduce an earlier package of assumptions but diverge from one another in ways consistent with separate development.

No individual cue is decisive. Citation can establish awareness without proving dependence; absence of citation can reflect convention, memory, strategic omission, or indirect learning. Social proximity can make transmission possible without showing that it occurred. Close wording can be inherited from a textbook rather than from the apparent predecessor. Even chronological priority is weaker than it appears when dates refer to private notes, presentations, preprints, editions, translations, and surviving copies. The system should therefore expose an evidential matrix and the rival explanations it supports rather than collapsing heterogeneous observations into a single edge weight.

This framing makes historical reconstruction resemble causal inference under severe missing data, but the analogy has limits. There is rarely a repeatable intervention that settles intellectual influence, and the relevant variables are themselves interpreted. The practical response is triangulation: semantic alignment, citation and social networks, archival evidence, version history, and expert contextual reading should constrain one another. The atlas becomes valuable not by eliminating historical judgment but by showing which conclusion depends on which evidence and which alternative remains live.

Uncertainty must be typed here as well. There is uncertainty from missing documents, uncertainty from ambiguous language, uncertainty from competing decompositions, uncertainty from model error, and uncertainty about historical causation. A single probability of "same idea" cannot communicate these sources. A lineage judgment may have high structural confidence and low influence confidence. It may be temporally possible but documentary weak. It may depend heavily on one translation. These distinctions should shape both the interface and the evaluation.

The deepest implication is that identity and history cannot be cleanly separated. What counts as the same idea affects which sources appear to be predecessors; the discovered predecessors can in turn change the representation of the idea. The process is iterative. An initial abstraction guides retrieval, retrieved sources reveal variants, and those variants force the abstraction to be revised. A neuro-symbolic atlas should support this hermeneutic loop while making each revision traceable. The goal is not to eliminate interpretation, but to transform interpretation from an invisible precondition into an explicit and testable component of the system.

4 Novelty, Priority, and Scientific Value

4.1 Novelty is a relation to a dated and bounded knowledge state

Novelty is not an intrinsic property carried by an idea in isolation. An idea is new relative to what was known, accessible, and representable at a time. The same contribution can be novel to a field, familiar in another discipline, independently rediscovered by a group that lacked access to the precedent, or old in mechanism but new in evidence and capability. Any automated judgment that omits the reference class will convert a relational concept into a misleading label.

The reference state should be written as K_t, a representation of prior knowledge at time t. This notation immediately raises questions that ordinary novelty scores suppress. Which languages, source types, and archives are included? Does K_t contain only published material or also preprints, patents, code, conference talks, and correspondence? How are inaccessible or undigitised sources represented? Which conceptual vocabulary and level of abstraction are used? A novelty claim is only as meaningful as its answer to these questions.

A structured conception treats novelty as resistance to reconstruction from K_t. Suppose a candidate idea I is compared with sets of prior components S and with admissible transformations T drawn from a grammar T_Q. One possible research objective is

$$R_Q\left(I|K_t\right) = \min_{S\subseteq K_t,\, T\in T_Q} \left[C(S,T) + D_Q(I,T(S))\right].$$

The term $C(S,T)$ measures the complexity of selecting and transforming prior components; D_Q measures the remaining mismatch under the question-specific representation. A low value means that the candidate can be economically reconstructed from known material. A high value means that the best available reconstruction is costly or leaves substantial unexplained structure. This formulation draws inspiration from minimum-description-length reasoning (Rissanen 1978; Grünwald 2007), but it should not be interpreted as a complete definition of scientific novelty. Compressibility depends on the code, the representation, and the available components. A random or incoherent object can also be difficult to compress.

The scientifically useful output is not the scalar R_Q alone. It is the reconstruction itself and the location of its residuals. The system should show which goals, mechanisms, assumptions, relations, evidential forms, or capabilities are inherited and which remain unexplained. It should then test the stability of those residuals under alternative decompositions, larger corpora, and different transformation grammars. A novelty claim that disappears when one adjacent field is added to the corpus is field-relative, not necessarily false. A residual that survives multiple representations is more interesting, but still requires interpretation.

Figure 3 makes the logic explicit. The candidate, prior knowledge state, and transformation grammar jointly determine the best reconstruction. What remains is a role-specific residual. The diagram then branches into four competing interpretations: substantive novelty, missing precedent, representational mismatch, and extraction error or incoherence. The same numerical residual is compatible with all four. Scientific validity depends on separating them.

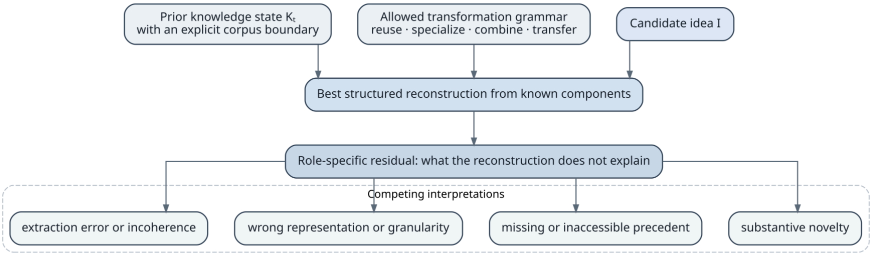

Figure 3. Structured novelty inference. A residual becomes evidence of novelty only after alternatives involving missing knowledge, poor representation, and error have been examined.

This approach yields a multidimensional novelty profile. A contribution may introduce a new problem, a new mechanism, a new combination, a transfer to a new domain, a new evidential technique, or a capability that was previously unavailable. These dimensions should not be collapsed prematurely. The difference between a new mechanism and a first effective implementation matters for scientific understanding and for credit. It also matters for AI training: the model should learn that "not previously demonstrated" and "not previously conceived" are different claims.

Table 5 identifies common sources of apparent novelty and the interpretation they require. The table is designed as a diagnostic aid. It shows why high semantic distance or lack of a retrieved match cannot serve as a verdict.

Table 5. Apparent novelty has several causes that must be distinguished before a contribution can be characterised.

Source of apparent novelty	Required interpretation
New terminology for an established structure	Test paraphrase and role alignment across vocabularies before inferring conceptual change.
Transfer of a known mechanism to a new domain	Separate mechanistic inheritance from novelty in assumptions, integration, evidence, or capability.

Source of apparent novelty	Required interpretation
Unusual recombination of familiar components	Analyse whether the organisation creates a new causal or functional property rather than rewarding rarity alone.
Finer implementation detail	Determine whether the detail changes performance, validity, or capability at the resolution relevant to the inquiry.
Missing retrieval result	Examine corpus coverage, language, indexing, and representation before treating absence as evidence.
Contradictory or incoherent proposal	Difficulty of reconstruction may reflect error rather than creativity; scientific constraints must be applied.
First persuasive validation	Distinguish novelty of evidence and practical realisation from novelty of the underlying principle.
Removal of a restrictive assumption	A local structural change may have major value even when most components are inherited.
Newly available enabling technology	Credit may belong partly to implementation or infrastructure rather than to a previously articulated mechanism.

The table directs attention from rarity to transformation. Scientometric research on atypical combinations is informative at scale, but unusual co-occurrence is only a proxy for conceptual novelty (Uzzi et al. 2013; Fortunato et al. 2018). A rare combination can be arbitrary; a common combination can become scientifically decisive when one assumption is changed or one relation is reorganised. The atlas should use population-level signals to guide retrieval, not to replace mechanism-level analysis.

Recent work on automated novelty assessment reinforces this caution. Literature-grounded systems improve on unguided model judgments by retrieving and comparing earlier work (Shahid et al. 2025). Yet benchmarks show that persuasive reasoning does not guarantee agreement with experts (Schopf and Färber 2026). The open-world character of novelty means that no benchmark can certify global absence of precedent. Evaluation must therefore measure evidence coverage, calibration, and robustness to corpus expansion in addition to agreement with a static label.

4.2 Priority is inference under incomplete and socially structured evidence

The desire to identify the “first appearance” of an idea is understandable. Priority structures citation, reputation, historical narrative, and sometimes legal or commercial claims. It is also one of the places where an atlas could do the most harm if it presents a clean answer unsupported by the record. The scientifically defensible target is narrower: the earliest known attestation within an explicitly described corpus and interpretation.

Let $A_C(I)$ denote the verified attestations of an operational idea I found within corpus C. The earliest known attestation is

$$a^*(I, C) = arg\,min_{a \in A_C(I)} date(a).$$

This equation is simple, but every term is contested. An attestation must be judged relevant under an identity model. The date may refer to composition, circulation, presentation, publication, or a later surviving copy. The corpus may exclude languages, archives, oral traditions, or proprietary records. Even the apparent minimum can change when a source is redated or newly discovered. The output must therefore report C, the dating evidence, the identity relation used, and unresolved alternatives.

Priority should be decomposed into several events that are often conflated. The first surviving formulation is not necessarily the first conception. The first public disclosure may precede formal publication. The first mathematical proof may follow an informal insight. The first working implementation may reveal constraints absent from earlier proposals. A later author may independently rediscover the mechanism and supply the first evidence that makes it scientifically consequential. A one-field “inventor” attribution suppresses this structure.

Historical evidence is censored in systematic ways. Documents are lost, archives remain closed, publications in dominant languages are more heavily indexed, and the contributions of less prestigious institutions are less likely to be cited and preserved. The Matthew effect amplifies visibility once recognition has accumulated (Merton 1968). A computational system trained on existing citation patterns can reproduce this inequality while appearing objective. Priority analysis must therefore include a model of coverage and a distinction between “not found” and “known not to exist.”

The system should represent priority claims as arguments. An argument links an identity judgment to one or more dated attestations, records the sources searched, and states why competing candidates were excluded. It also records challenges: an earlier source, an alternative translation, a claim that the abstraction is too broad, or evidence of independent development. The result is not a permanent label but a versioned historiographical object.

Influence and priority must remain separate. An earlier attestation can establish chronological precedence without showing that later researchers knew it. Conversely, a later textbook or intermediary may have been the actual source of influence even though it was not the first formulation. The atlas should allow a lineage to pass through transmission nodes that are not conceptually original. This distinction is valuable because science advances through both invention and diffusion.

Legal concepts such as patent novelty and inventive step provide useful analogies but should not be imported uncritically. Patent systems evaluate claims under jurisdiction-specific doctrines and evidential standards. Scientific originality includes explanatory, methodological, and evidential contributions that do not map cleanly onto patentability. An atlas could support prior-art search, but its scientific ontology should not be determined by legal categories.

A mature priority model would assign uncertainty for several reasons. It could express uncertainty that the source contains the same idea, uncertainty in dating, uncertainty in corpus coverage, and uncertainty about transmission. These should remain separate because they demand different responses. A semantic dispute requires expert interpretation; a dating dispute requires archival work; a coverage problem requires broader search; an influence dispute requires documentary and social evidence.

The value of such a system is not that it will end priority disputes. It can make them more productive by exposing where the disagreement lies. It can also reduce the ease with which polished recent formulations erase older or parallel traditions. The ethical standard should be conservative: claims about earliest appearance and credit require stronger evidence than claims about structural similarity.

4.3 Useful novelty and epistemic gain

Originality alone is not scientific value. Creativity research commonly pairs originality with effectiveness or appropriateness (Runco and Jaeger 2012; Boden 2004). In science, usefulness is plural. An idea can be valuable because it explains more with less, predicts accurately, enables control, makes a difficult measurement possible, opens a new experimental regime, reduces cost, unifies separate domains, reveals a boundary condition, or generates further productive questions. These achievements are not reducible to citation count and may emerge long after publication.

Novelty and value should therefore be represented by separate profiles. A highly original idea may be untestable, incoherent, or irrelevant. A modest modification may remove a bottleneck and transform practice. Stokes's analysis of use-inspired basic research illustrates that fundamental understanding and practical use are not opposite ends of one scale (Stokes 1997). The atlas should preserve such multidimensionality rather than rank every idea on one axis.

One family of value measures concerns epistemic gain. An idea may reduce uncertainty, discriminate between theories, compress a set of observations, or reveal which experiment would be most informative. Another concerns capability: it may make a prediction, intervention, computation, or observation possible that was not feasible before. A third concerns robustness and scope: the idea may continue to work under broader conditions or fail more predictably. A fourth concerns generativity: it may supply a reusable representation or mechanism from which further work can be derived.

These dimensions interact with evidence. Theoretical promise should not be scored as established utility, and early negative results should not automatically erase long-term value. The system needs temporal states such as conjectured, demonstrated under limited conditions, replicated, generalised, contradicted, and superseded. It should also represent the cost and risk of obtaining further evidence. A high-potential idea with expensive or hazardous tests occupies a different position from a cheap, immediately testable one.

Prediction of future usefulness is intrinsically difficult because value depends on social and technological contingencies. A mathematical technique may wait decades for suitable computation. A mechanism may become important only after a measurement instrument is invented. Institutional attention can create feedback loops in which funded ideas acquire evidence and unfunded ones remain speculative. The atlas should therefore avoid training a utility predictor solely on past recognition. That would turn historical advantage into a normative rule.

The science of science shows that research systems can be biased against novelty and that researchers balance risky innovation with conventional strategies (Foster, Rzhetsky, and Evans 2015; Wang, Veugelers, and Stephan 2017). An atlas could either mitigate or intensify this tendency. It could reveal overlooked precedents and unexpected connections, broadening the search space. It could also become a gatekeeping instrument that penalises ideas because they do not resemble recognised successful trajectories. Governance must therefore separate exploratory support from high-stakes evaluation.

A useful design principle is Pareto presentation. Instead of computing one total score, the system displays how a proposal lies across novelty, evidence, expected information gain, feasibility, scope, cost, and risk. Different users can then apply explicit priorities. Peer reviewers may focus on evidential support and contribution relative to the literature; a research programme may accept lower feasibility for high information gain; an engineering team may prioritise robustness and cost. The atlas supplies structured comparison without pretending that one value function governs science.

Ex ante and ex post value must also be distinguished. Before an experiment, a hypothesis may be valuable because it partitions the space of possibilities and supports a decisive test, even if it later proves false. After the experiment, the same hypothesis may be remembered only as an error. Conversely, a result that appears minor at publication can become enabling when another technology changes the surrounding possibility space. A scientific atlas should preserve the decision context in which an idea was proposed: what was known, which alternatives were live, what evidence was affordable, and what outcome the inquiry could discriminate.

This makes the value of negative results conceptually central. A failed intervention can eliminate a mechanism, expose a hidden assumption, or identify a boundary beyond which an analogy breaks. Such contributions may have low novelty in their components and high epistemic value in the constraint they establish. Counterexamples, impossibility results, calibration failures, and replications that narrow an effect are not merely deficient successes. They reshape the space in which later ideas can be evaluated. An atlas focused only on positive capabilities would systematically misrepresent scientific progress.

Expected information gain offers one formal lens, but it cannot serve as a universal value measure. Its result depends on the prior distribution, the question, and the available action set. A cheap experiment that sharply updates a local model may have greater immediate information value than a visionary theory whose consequences cannot yet be observed. The atlas can record these dependencies and compare alternative research moves, but it should not conceal normative choices inside an apparently objective calculation.

Value is therefore relational in a second sense: it is value for a community, problem, and historical state. A technique may be redundant in a well-equipped laboratory and transformative in a resource-constrained setting. A representation may have modest predictive gain but enormous educational or integrative value. Preserving these contexts is necessary both for fairness and for scientific accuracy. Without them, the system would learn that impact is an intrinsic property of ideas rather than an interaction between ideas and the worlds into which they arrive.

The phrase “new, but not new for its own sake” can now be stated more precisely. The target is a transformation of prior knowledge that produces epistemic or operational gain. The transformation may be radical or local. What matters is whether it changes what can be explained, predicted, tested, built, or connected, and whether that change survives critical comparison with the prior record. A working atlas would not automate this judgment. It would make the judgment accountable to structure, evidence, and alternatives.

5 The Neuro-Symbolic Epistemic Machine

5.1 Representation must be plural, evidence-grounded, and revisable

The proposed atlas is naturally described as neuro-symbolic, but the term can conceal more than it clarifies. Many systems are called neuro-symbolic because they combine a neural network with a knowledge graph or because they translate model outputs into logical predicates. The scientific requirement here is stricter. The architecture must allocate different epistemic responsibilities to different representational forms and prevent one layer from silently overriding the commitments of another.

Neural models are well suited to open-ended linguistic variation. They can retrieve semantically related passages, align terminology across fields, propose role structures, translate historical language, and generate candidate analogies. Their weakness is not merely hallucination. Distributed representations make it difficult to determine which source supports a particular abstraction, which invariant produced a match, and whether an answer remains valid after the knowledge base changes. A fluent model can produce a coherent lineage story without having distinguished resemblance from influence.

Symbolic representations are suited to explicit roles, typed relations, constraints, and inspectable inference paths. They can state that a later method preserves a goal, replaces a mechanism, removes an assumption, and imports an evaluation technique from a third source. They can enforce temporal constraints and prevent an earlier work from inheriting from a later one. Their weakness is brittleness and the cost of constructing the symbols. They do not solve lexical variation or open-ended abstraction simply by being explicit.

Probabilistic representations are required because many relevant claims are uncertain rather than simply true or false. The uncertainty may concern extraction, interpretation, dating, influence, or corpus coverage. Probabilistic logic and statistical relational learning offer ways to combine uncertain evidence with relational constraints (Manhaeve et al. 2018; Raedt et al. 2020). Yet probability alone is not enough. Competing decompositions can be qualitatively different hypotheses, and a single confidence value may hide the source of disagreement.

The architecture should therefore preserve four kinds of object. Documentary objects identify source material. Interpretive objects represent proposed operational ideas. Relational objects state typed identity, transformation, support, and lineage claims. Argument objects connect those claims to evidence, rules, model versions, and challenges. This design draws on semantic publishing, nanopublications, and micropublications, which demonstrate that small claims and their provenance can be made machine-actionable (Groth, Gibson, and Velterop 2010; Kuhn et al. 2018; Clark, Ciccarese, and Goble 2014). The atlas extends this discipline to structured ideas and historical comparison.

Evidence grounding must be local. A paper-level citation is insufficient when the system claims that a specific mechanism appears in an earlier source. The relation should point to the passage, equation, diagram, code segment, or experimental artefact that justifies the abstraction. If the claim is a synthesis not stated by any source, it must be labelled as such and linked to the evidence from which it was inferred. This distinction is central for trust: the user must be able to tell what was observed, what was interpreted, and what was generated.

Representation must also be plural. A symbolic graph should not enforce one irreversible decomposition. It should allow several views of the same evidence and relations between those views. One representation may be mechanism-centred, another functional, another historical. The graph can record that a coarse object abstracts a finer one, that two experts disagree about a boundary, or that a model's candidate interpretation was rejected. Plurality is not database disorder when it is typed and versioned.

Table 6 summarises the epistemic division of labour. The second column describes both the distinctive contribution of each layer and the boundary it should not cross without additional support.

Table 6. A neuro-symbolic atlas requires an explicit division of epistemic labour among representational layers.

Representational layer	Epistemic responsibility and limit
Source and attestation layer	Preserves the documentary record, exact locations, versions, and artefacts; it does not by itself determine the identity of the represented idea.
Neural semantic layer	Retrieves, translates, aligns, abstracts, and proposes candidate structures; its outputs remain hypotheses until grounded and adjudicated.
Symbolic structural layer	Records roles, typed relations, constraints, and explicit transformation paths; it should not pretend that its chosen ontology is complete or uniquely correct.
Probabilistic and temporal layer	Represents uncertainty, competing hypotheses, censoring, and time-indexed knowledge states; probability must remain attached to the source of uncertainty.

Representational layer	Epistemic responsibility and limit
Argument and provenance layer	Connects conclusions to evidence, agents, models, rules, revisions, and objections; it prevents synthesis from being confused with quotation.
Human interpretive layer	Resolves difficult cases, supplies domain and historical judgment, and challenges the system; expertise is recorded rather than treated as infallible authority.

The table explains why a single end-to-end model is not the target, even if such a model could achieve high benchmark accuracy. The atlas is intended to support claims that can affect credit, review, and research direction. Those claims must remain auditable after the model is replaced. Representational separation gives the system institutional memory: evidence and arguments survive changes in the neural component.

5.2 Neural proposal, symbolic adjudication, and revisable interpretation

The architecture should be organised around proposal and adjudication rather than extraction and storage. “Extraction” suggests that the correct idea structure is already present in the text waiting to be copied. In reality, the system proposes a representation by combining spans, background knowledge, and a task model. Adjudication evaluates whether the proposal is supported, coherent, and appropriate for the inquiry.

A neural proposal begins with source selection. Retrieval must operate over multiple views: lexical, semantic, citation-based, temporal, and structural. A candidate precedent can be relevant because it uses the same terminology, because it instantiates the same mechanism under different language, or because it occupies a key position in a citation lineage. The retrieval model should preserve which path produced the candidate. Otherwise, downstream users cannot distinguish direct evidence from exploratory analogy.

The model then proposes roles and relations. It may identify a goal, mechanism, assumptions, and outcomes; align them with existing idea objects; and suggest a transformation such as specialisation or removal of a limitation. Crucially, it should produce alternatives when the text supports more than one decomposition. A calibrated system can state that a passage probably instantiates either a general optimisation principle or a domain-specific procedure and that the difference affects the novelty analysis.

Symbolic adjudication checks these proposals against constraints. Some constraints are logical: a relation may require compatible object types. Some are temporal: an influence edge cannot run backward in time. Some are causal or mechanistic: claimed equivalence may require preservation of dependency structure. Some are evidential: an assertion of influence requires stronger support than a claim of similarity. Some are historiographical: a translation may not justify attributing modern terminology to an earlier author.

Constraint checking should not be restricted to rejection. It can generate questions that guide further retrieval. If two methods appear mechanistically equivalent but have incompatible assumptions, the system searches for a finer representation or evidence that one assumption is dispensable. If an influence hypothesis is semantically strong but documentary weak, the system searches for intermediary publications, correspondence, shared collaborators, or educational channels. Reasoning becomes an active process of evidence acquisition.

Probabilistic interpretation combines uncertain evidence without erasing its type. A useful model might assign separate distributions to extraction correctness, structural alignment, historical influence, and corpus completeness. The final report can then state, for example, that mechanistic equivalence is strongly supported, influence is possible but undocumented, and priority remains unstable because non-English archival coverage is poor. This is more informative than a single confidence score of 0.72.

Neuro-symbolic systems such as DeepProbLog and Logic Tensor Networks illustrate different ways of combining learning with formal constraints (Manhaeve et al. 2018; Badreddine et al. 2022). The atlas is not committed to one framework. Its conceptual requirement is that the inference path remain inspectable and that uncertainty be attached to claims. Some components may use probabilistic logic, others differentiable constraints, graph neural networks, or learned theorem-guided retrieval. The architecture should be judged by epistemic behaviour rather than by membership in a brand of neuro-symbolic AI (Garcez and Lamb 2020; Hitzler and Sarker 2022; Sarker et al. 2021).

Figure 4 depicts the system as a cycle. Evidence feeds neural proposal; candidate assertions enter symbolic adjudication; probabilistic and temporal reasoning assesses competing hypotheses; human challenge introduces corrections and new sources; and those revisions alter the evidence and learning process. The cycle has no final "truth database." It produces a succession of better-supported and more precisely qualified interpretations.

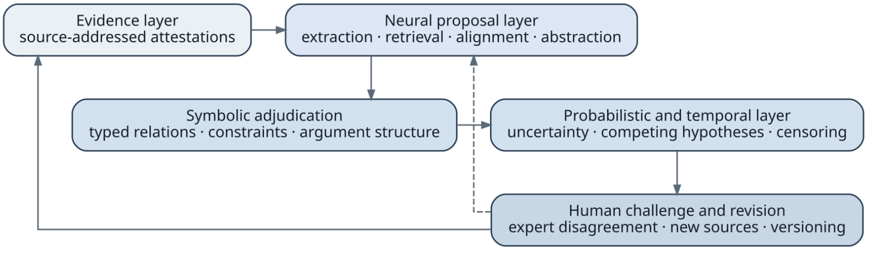

Figure 4. The neuro-symbolic epistemic cycle. Neural flexibility is constrained by symbolic structure, uncertainty is represented explicitly, and human challenge can revise both evidence and models.

The human role in the figure is not a residual task left over because automation is imperfect. Some questions are interpretive in principle. Experts decide which differences are explanatory, whether a historical source performs the same intellectual act, and how much abstraction a comparison can tolerate. The system can make these judgments more consistent and evidence-rich, but it should not hide them behind automation. Human decisions themselves become provenance-bearing objects that can be studied for disagreement and bias.

An atlas of ideas must expect its conclusions to change. Classical databases often assume that adding facts monotonically increases knowledge. Historical and semantic interpretation is non-monotonic: a new source can invalidate a priority claim, a better translation can split an apparent identity, and a new experiment can transform a mechanism previously treated as equivalent. The architecture should therefore treat revision as normal rather than as corruption.

Versioning must operate at several levels. Sources have editions and corrections. Attestations can be relocated when pagination changes. Idea objects can be refined or split. Relations can be accepted, challenged, or deprecated. Ontology terms can change. Model outputs depend on model versions and prompts. A user examining a claim should be able to reconstruct the state of the atlas at a previous date and understand why the current view differs.

Argumentation provides a natural structure for revision. A claim such as "work B independently rediscovered the mechanism in work A" can be supported by structural alignment, temporal separation, and absence of plausible transmission, while challenged by evidence of a shared intermediary. The system need not force every argument into deductive logic. It can represent defeasible rules, evidential weights, and unresolved rebuttals. What matters is that the conclusion is not detached from its reasons.

This approach also reduces the temptation to encode governance decisions as technical defaults. Consider the threshold for merging two idea objects. A low threshold creates fragmented duplicates; a high threshold erases meaningful variation. The choice affects attribution and novelty. It should be visible, testable, and, where appropriate, adjustable by domain. Likewise, the decision to treat a source as authoritative, a language model as a curator, or expert consensus as sufficient is a governance choice with epistemic consequences.

The atlas should support contestability. Researchers need a way to challenge an identity relation, supply an earlier attestation, propose an alternative decomposition, or flag a translation problem. Challenges should not immediately overwrite accepted records; they should create parallel argument states until adjudicated. This prevents both vandalism and premature closure. It also makes the atlas a site of scholarship rather than a static reference work.

Bias audits must examine representation and coverage, not only model outputs. A system can be calibrated on its indexed corpus and still be systematically unjust because the corpus excludes relevant traditions. It can accurately reproduce citation patterns that reflect historical inequality. Coverage metadata should therefore be part of every high-stakes novelty or priority analysis. The absence of accessible sources in a region, language, or publication type should lower confidence and trigger targeted search.

Open standards such as PROV-O and FAIR principles provide part of the infrastructure for traceability and reuse (World Wide Web Consortium 2013; Wilkinson et al. 2016). The deeper governance challenge is semantic. Who may create or revise idea families? How are minority interpretations preserved? When is a relation sufficiently supported for high-stakes use? Which claims should remain advisory? These questions require institutional design, but the conceptual answer is clear: no single opaque model should control identity, novelty, and credit.

The architecture should also distinguish exploration from adjudication. During exploration, the system can generate speculative analogies, distant candidates, and alternative decompositions with permissive thresholds. During adjudication, evidential standards become stricter, and unsupported relations remain hypotheses. Mixing the two modes creates danger. A creative suggestion is useful precisely because it can be weakly supported; a priority judgment is credible only when it is conservatively supported.

The proposed neuro-symbolic machine is therefore not a compromise between "neural" and "symbolic" camps. It is an epistemic constitution. It assigns flexibility, explicitness, uncertainty, and authority to different layers and defines how they may correct one another. Its success depends less on whether every component is differentiable than on whether the whole system makes scientific interpretation inspectable, revisable, and resistant to the confident collapse of distinct relations.

6 VSA/HDC as a Theory-Driven Experiment in Identity and Novelty

6.1 Why high-dimensional symbolic vectors are relevant

Vector Symbolic Architectures and Hyperdimensional Computing describe a family of models that encode structured information in high-dimensional distributed vectors. Their characteristic operations bind items into role–filler associations, bundle several associations into a composite representation, and often permute or transform vectors to represent order and position. Because unrelated high-dimensional vectors are approximately dissimilar and because superposed structures can be recovered approximately, these models occupy an unusual position between classical symbols and learned embeddings (Kanerva 2009; Kleyko et al. 2023).

This position makes VSA/HDC relevant to the atlas for conceptual rather than merely computational reasons. Idea identity requires both structure and tolerance. A representation must preserve that a mechanism fills one role and an assumption another, while recognising approximate correspondences across terminology and domains. Classical symbolic structures preserve roles exactly but require explicit matching. Ordinary embeddings provide flexible similarity but tend to entangle roles. VSA/HDC offers a hypothesis about how role-sensitive structures could inhabit a distributed similarity space.

The family is diverse. Tensor product representations preserve compositional structure in a high-dimensional tensor product (Smolensky 1990). Holographic Reduced Representations compress binding through circular convolution (Plate 1995). Binary spatter codes, multiply–add–permute models, sparse distributed representations, and other architectures use different vector spaces and operations. Their algebraic properties are not interchangeable. Binding may be self-inverse in one architecture, require an explicit inverse in another, or preserve similarity to different degrees. Comparative work shows that the choice of architecture affects analogical reasoning, capacity, and decoding (Schlegel, Neubert, and Protzel 2022; Kleyko et al. 2023). Any proposal that speaks of "the VSA representation" without specifying the algebra is underdefined.

The attraction of high-dimensional vectors lies partly in concentration phenomena. Random vectors in a sufficiently high-dimensional space are likely to be nearly orthogonal, allowing many symbols to be assigned without careful geometric placement. Bundling creates a noisy superposition whose constituents remain detectable by similarity. Binding produces a vector dissimilar to its inputs while permitting approximate recovery when one factor is known. These operations can encode sets, sequences, graphs, and role–filler structures in fixed-size distributed representations.

For an atlas, fixed size is less important than compositional comparison. Suppose two ideas share a goal and mechanism but differ in assumptions and evidence. A role-bound representation can in principle recover each role separately, estimate correspondence, and still compare the whole. The representation can also support associative retrieval: a partial query specifying a problem and mechanism may retrieve candidates whose surface language differs. This is a stronger use of VSA/HDC than employing hypervectors as efficient classifiers.

The approach also suggests a computational treatment of transformations. If related ideas differ through operations such as substitution, specialisation, role transfer, or recombination, those operations may themselves be represented and compared. Recurrent transformation patterns could become first-class objects: "remove an independence assumption," "replace deterministic control with feedback," or "transfer a mechanism from a discrete to a continuous domain." An atlas could then search not only for similar ideas but for similar *changes*.

There is, however, a danger of mistaking representational convenience for semantic theory. High dimensionality does not determine which roles should exist, which fillers are equivalent, or which transformations are scientifically meaningful. These choices come from the operational ontology, evidence, and task. Random codebooks provide separability, not interpretation. Learned codebooks may capture useful regularities, but they can also reproduce biases and entangle distinctions the symbolic layer needs to preserve.

VSA/HDC should therefore be tested as an associative compositional memory within a larger neuro-symbolic system. The authoritative record remains the evidence-grounded graph. Hypervectors provide rapid retrieval, approximate alignment, transformation search, and factorisation hypotheses. Their outputs must be translated back into role-level assertions and checked against sources. This division of labour is represented in Figure 5.

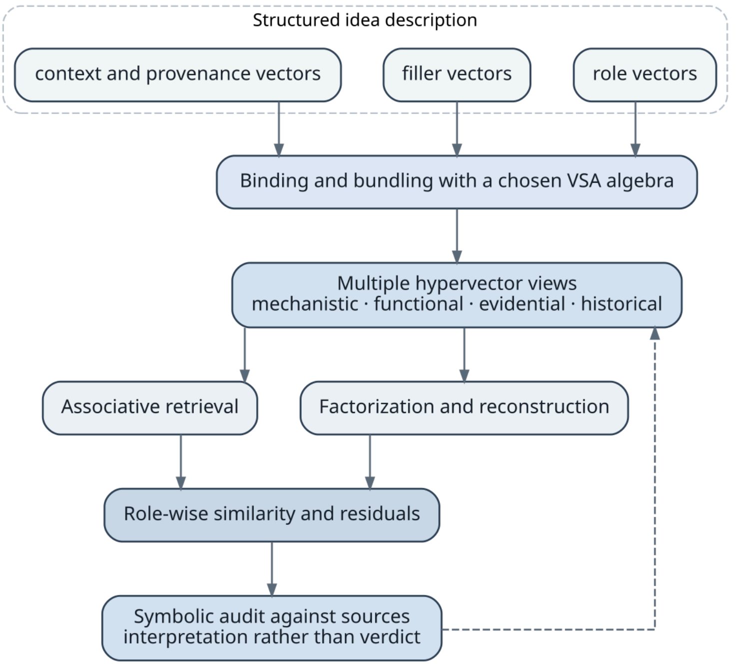

Figure 5. A proposed role for VSA/HDC. Structured descriptions are encoded into several hypervector views, used for retrieval and reconstruction, and then audited against the symbolic and documentary layers.

The figure emphasises two design choices. First, there is not one total hypervector. Mechanistic, functional, evidential, and historical views may use different role sets and codebooks. Second, the path ends in symbolic audit. A role-wise residual is a clue about where an alignment fails, not a direct declaration of novelty.

6.2 Role–filler binding, reconstruction, and the novelty residual

Let R_r denote a hypervector for role r and $F_r(I)$ a hypervector representing the filler of that role in idea object I. Using an abstract binding operator $\odot$ and bundling operator $\oplus$, a simple representation is

$$H(I) = \oplus_{r \in R}\left(R_r \odot F_r(I)\right).$$

The notation deliberately abstracts over the VSA family. In an HRR-like model, binding may use circular convolution and bundling vector addition. In a binary architecture, binding and bundling may use XOR and majority operations. The scientific claim is not tied to one choice. It is that binding can preserve the association between a role and its filler inside a distributed composite.

The role set R should reflect the operational representation. It may include goal, mechanism, assumptions, context, outcome, evidence, and provenance, but each can be decomposed. A mechanism view might bind entities, activities, temporal relations, feedback paths, and causal dependencies. An evidential view might bind claim type, study design, measurement, result, and replication status. A lineage view might bind inherited components, transformations, source attestations, and dates.

A single bundled vector can become overloaded when many roles and nested structures are superposed. Multi-view representation addresses this problem and respects ontological pluralism. For each view v, define

$$H_v(I) = \oplus_{r \in R_v} \left(R_{v,r} \odot F_{v,r}(I) \right).$$

The idea object is then represented by a collection $\{H_v(I)\}_{v \in V}$ rather than by one universal vector. The views may share some atomic code vectors while preserving different bindings and resolutions. This makes query-relative identity natural: a mechanistic comparison weights the mechanism view; a priority inquiry emphasises historical and provenance views; an analogy query may compare relational structure while suppressing domain-specific fillers.

Multi-view encoding also mitigates a conceptual error common in embedding systems. If all information is compressed into one geometry, proximity on one dimension can dominate another. Two works with similar language and citations may appear close even when their mechanisms differ. Separate views permit late combination under an explicit question. The system can report that linguistic and functional similarity are high while mechanistic similarity is low.

Nested structure requires further operations. Order can be represented through permutation, position vectors, or recursive binding. Graph relations can be encoded by binding source, relation, and target roles. For a mechanism with feedback, the representation must distinguish a set of components from the organised cycle among them. VSA/HDC provides several constructions for sequences and graphs, but their capacity and similarity behaviour differ. The atlas should compare architectures empirically rather than assume that one encoding preserves every relevant relation (Kleyko et al. 2023; Schlegel, Neubert, and Protzel 2022).

Recovering a role filler from a composite usually involves unbinding. In abstract form, the estimate of the filler for role r is

$$\widetilde{F}_r(I) = R_r^{-1} \odot H(I),$$

where R_r^{-1} denotes the architecture-appropriate inverse or approximate inverse. Because bundling introduces cross-talk, $\widetilde{F}_r(I)$ is noisy. A cleanup memory compares it with a codebook of known fillers or candidate structures. This mechanism is useful for interpretation: a system can ask what mechanism is encoded in a retrieved idea or whether two candidates fill a role with compatible structures.

The cleanup step is also a source of epistemic risk. If the correct filler is absent from the codebook, the memory will return the nearest known item, making genuine novelty look like a distorted instance of the familiar. If the codebook is excessively large or poorly separated, recovery fails. If fillers are generated by a neural encoder rather than assigned randomly, learned geometry can help semantic generalisation but may reduce the quasi-orthogonality on which VSA operations depend. The interaction between learned semantic vectors and algebraic codebooks is a central research problem.

One possibility is a dual representation. Atomic symbols and stable ontology roles receive controlled code vectors; open-ended content receives learned vectors projected into the hyperdimensional space; and the link to the original source is preserved symbolically. Another is to learn codebooks under constraints that maintain binding recoverability and role separation. The relevant objective is not ordinary downstream accuracy alone. It must measure whether composition, unbinding, and factorisation remain reliable under the scale and ambiguity of scientific concepts.

Representation should include uncertainty. A role may have several candidate fillers with different weights, or its content may be partially unknown. Bundling can encode weighted alternatives, but the resulting vector should not be mistaken for a probability distribution without calibration. A better design stores the alternative symbolic hypotheses and uses hypervectors to retrieve and compare them. The VSA layer can represent a superposition of possibilities; the probabilistic layer records what that superposition means.

Provenance requires special care. It is tempting to bind source identifiers directly into the idea vector. Doing so can improve retrieval of historically related items but contaminate semantic comparison: two identical mechanisms from different sources should not appear dissimilar merely because their provenance differs. Multi-view design solves this by keeping provenance and semantic structure separate, combining them only for questions that require both. This separation is another reason to reject the idea of a single canonical hypervector.

The most distinctive VSA/HDC research direction concerns reconstructability. If an idea is composed of known role–filler structures and transformations, its hypervector should be approximable by combining the corresponding prior hypervectors. A candidate that cannot be reconstructed may contain a new component or organisation. This suggests a computational proxy for one dimension of novelty.

The first step is candidate retrieval. Given $H_v(I)$, associative memory identifies earlier objects with similar functional, mechanistic, or contextual views. The second step searches for combinations and transformations that improve the match. The third attempts to factor the candidate into elements drawn from codebooks of prior components and transformation operators. Resonator networks were developed precisely to solve factorisation problems in distributed compositional representations by alternating VSA operations with associative cleanup (Frady et al. 2020; Kent et al. 2020). Their performance makes them relevant to a search over possible antecedent structures.

A reconstruction model might seek a prior component set S and transformation sequence T whose encoded result is close to the candidate. Rather than treating the global difference as novelty, it should unbind corresponding roles and compute role-specific residuals. Let $\hat{I} = T(S)$ denote the best reconstructed structure. For each role r, define

$$\rho_r\left(I, \hat{I}\right) = 1 - sim\left(\widetilde{F}_r(I), \widetilde{F}_r\left(\hat{I}\right)\right).$$

A high ρ_r indicates that the reconstruction does not explain the filler associated with role r. The residual profile $\rho\left(I, \hat{I}\right)$ is more informative than a single distance. It may show that the goal and evaluation are inherited, the mechanism is partly transformed, and one assumption has no close precedent.

The central scientific hypothesis can now be stated carefully: *under an adequate representation, codebook, corpus, and transformation grammar, persistent role-specific reconstruction residuals correlate with expert judgments of substantive novelty*. Every qualification is necessary. The hypothesis is empirical and may fail. It does not identify novelty with residual magnitude; it predicts a relationship after controlling for competing causes.

Factorisation faces combinatorial difficulty. An idea may draw components from many antecedents, and transformation sequences can proliferate. Resonator networks exploit superposition to search factor combinations, but they are not guaranteed to converge and have finite capacity (Kent et al. 2020). Scientific idea structures are also more open-ended than the controlled codebooks in many demonstrations. A realistic system needs strong retrieval priors, hierarchical search, and symbolic constraints to keep factorisation tractable.

The search space should include transformations as well as components. A candidate may be reconstructed from an earlier mechanism plus the operation "remove assumption A" and a measurement technique imported from another field. Transformation operators could be encoded as hypervectors or as symbolic programs whose effects are compiled into the VSA space. Recurrent transformations learned from curated lineages would constrain the search. The result would resemble a grammar of scientific change rather than a bag-of-ideas recombination engine.

Several residuals can arise without novelty. Missing literature is the most obvious. If the relevant precedent is absent from K_t, the reconstruction fails. A defective ontology can produce the same effect: the system may lack a role needed to align two formulations. Wrong granularity can split one component into many or collapse a decisive distinction. Neural extraction can misrepresent the candidate. Codebook capacity can be exceeded, producing cross-talk. The candidate itself may be incoherent. Each failure has a different diagnostic signature, but none is guaranteed to be separable from true novelty by the VSA layer alone.

Stability analysis is therefore mandatory. The system should repeat reconstruction under alternative views, dimensions, codebooks, corpus samples, and transformation grammars. A residual that moves unpredictably with random initialisation is not a robust novelty signal. A residual that disappears after a modest corpus expansion indicates coverage sensitivity. A residual localised to one role across architectures and confirmed by expert comparison is stronger evidence.

The MDL interpretation introduced earlier provides a conceptual bridge. Reconstruction cost reflects not only distance but the complexity of the antecedent explanation. A candidate exactly representable as a contrived combination of dozens of unrelated precedents may be less genuinely “anticipated” than one derived by a simple transformation from a single lineage. VSA factorisation can generate candidate decompositions, while a symbolic description language assigns interpretable costs. The hybrid system can prefer explanations that are both geometrically adequate and structurally economical.

This framework may also identify *negative novelty*: a claim presented as new whose structure is readily reconstructed from earlier work. The output should still avoid accusation. It should display the reconstruction, sources, and assumptions, allowing experts to judge whether the overlap is substantial. A low residual can coexist with novelty in evidence, implementation, or domain. The atlas analyses dimensions; it does not reduce scholarly credit to a vector distance.

6.3 Transformations, analogy, and the limits of the VSA hypothesis

Analogy is central to scientific discovery because mechanisms often migrate across domains. Structure-mapping theory distinguishes relational correspondence from mere attribute similarity (Gentner 1983; Gentner and Markman 1997). An atlas should retrieve cases that share an organisation of roles even when their entities and vocabulary differ. VSA/HDC is promising because binding can represent roles independently of fillers, allowing queries that suppress domain-specific content and compare relational structure.

Suppose a system encodes two mechanisms with different entities but similar feedback organisation. By unbinding entity fillers and retaining role relations, it can form an abstract structural view. Similarity in this view proposes an analogy. The symbolic layer then checks whether causal dependencies, temporal order, boundary conditions, and failure modes align. The result is not “these ideas are the same,” but “this relational mapping may support transfer under specified assumptions.”

Transformations can themselves be compared. If A becomes B through a change in role structure, and C becomes D through a comparable change, the system may identify an analogy between the two transitions. Some VSA architectures support operations that approximate such relational differences, but their behaviour depends on the binding algebra (Schlegel, Neubert, and Protzel 2022). A safer approach represents the transformation explicitly as a structured object and encodes that object in a dedicated view. Similarity between transformation hypervectors then guides retrieval, while symbolic alignment verifies the correspondence.

This capability could be more valuable than direct idea similarity. Researchers often seek a way to repair a limitation rather than a duplicate of their problem. An atlas might retrieve lineages in which an analogous assumption was removed, a feedback delay was handled, or a measurement bottleneck was overcome. The transfer remains speculative until domain constraints are checked, but the search is guided by the form of change rather than by shared terminology.

VSA/HDC also offers robustness to noisy and partial queries. A researcher may specify only a goal, a constraint, and a desired capability. Bundled representations can retrieve candidates matching subsets of those roles. Because similarity is distributed, the system can degrade gracefully when some information is missing. This is well suited to exploratory science, where the query itself may be incomplete.

The same properties create false positives. High-dimensional similarity can arise from shared common roles, and superposition can obscure which components produced the match. A ubiquitous pattern such as optimisation under constraints may connect many unrelated methods. Role weighting, inverse-document-frequency analogues for semantic components, and symbolic post-filtering are needed to prevent generic structures from dominating. Cross-domain analogy should be evaluated not only by inspiration but by structural fidelity and the rate of seductive but invalid transfers.

Capacity is a fundamental limit. Bundling too many associations increases noise. Deeply nested structures become difficult to decode. Factorisation requires finite codebooks and can converge to incorrect combinations. Larger dimensionality helps but does not eliminate these problems; it increases memory and computation and may postpone rather than solve open-world scaling. The relevant experiments must estimate capacity under realistic distributions of idea complexity, not only random synthetic vectors.

The codebook problem is equally fundamental. If atomic vectors are random, semantic similarity must be supplied through shared structure and learned mappings. If atomic vectors inherit neural geometry, binding operations may not have the desired algebraic properties. A hybrid codebook may work, but its stability across model versions and domains must be tested. The atlas cannot allow a change in the underlying language model to silently redefine historical identity.

Identifiability is a deeper problem than retrieval accuracy. Many combinations of role vectors, filler vectors, and binding conventions can produce similar global similarities. A successful match does not show that the recovered internal factors correspond to the scientifically intended decomposition. The system needs interventions on representation: hold the source evidence fixed, vary the codebook or algebra, and test whether the same role-level alignment survives. When several encodings are related by a harmless reparameterisation, conclusions should be invariant. When a conclusion changes because one architecture preserves order and another does not, the difference should be predicted by the theory rather than discovered as an unexplained benchmark fluctuation.

Learned codebooks introduce temporal drift. Updating a language encoder can move fillers, change neighbourhoods, and alter which historical objects are retrieved. One response is to anchor stable symbolic roles and canonical entities while learning only mappings for open-ended content. Another is to maintain explicit translation maps between codebook versions and to rerun calibration sets after every update. Either way, version identity becomes part of the provenance of a VSA judgment. A statement that two ideas were close under codebook V_3 is not automatically interchangeable with the same statement under V_7.

The VSA hypothesis must also be tested against alternatives designed for the same structural problem. Graph matching, tensor-product models, differentiable unification, probabilistic programs, and neural architectures with explicit slots may preserve role structure without relying on superposition and cleanup memory. VSA/HDC earns a place only if its particular combination of distributed robustness, algebraic binding, fixed-dimensional memory, and approximate factorisation produces advantages that these alternatives do not. Energy efficiency or hardware suitability may matter for deployment, but they do not establish semantic adequacy.

These constraints clarify the possible scientific contribution. The strongest result would not be that hypervectors can encode an example. It would be a theory connecting properties of the binding algebra, dimensionality, codebook design, and structural depth to specific errors in identity, analogy, and reconstruction. Such a theory could explain when distributed symbolic memory preserves the invariants needed by the atlas and when it necessarily loses them. Even a negative boundary theorem or a carefully mapped failure regime would advance the problem more than an attractive demonstration without identifiability.

Table 7 frames the principal VSA/HDC claims as falsifiable hypotheses. The second column states observations that would weaken or reject each claim. This table is important because the chapter is proposing a research direction, not endorsing a technology in advance.

Table 7. VSA/HDC should enter the project through explicit hypotheses and failure criteria.

Research hypothesis	Evidence that would weaken or falsify it
Role-bound hypervectors improve identity judgments beyond strong text and graph embeddings.	Gains disappear on contrastive tests where surface similarity and mechanistic identity are deliberately separated.
Multi-view encoding preserves useful distinctions better than one total embedding.	View separation adds complexity without improving calibration, transfer, or interpretability.
Factorisation recovers meaningful antecedent components and transformations.	Reconstructions are unstable across dimensions and codebooks, or experts judge recovered components to be post-hoc artefacts.
Role-specific residuals correlate with substantive novelty.	Residuals mainly track missing corpus coverage, extraction error, or unusual wording after appropriate controls.
Transformation representations support cross-domain analogical retrieval.	Retrieved analogies are no more structurally faithful or useful than strong relational graph and language-model baselines.
The VSA layer can scale while preserving auditability.	Capacity limits, cleanup errors, or codebook drift make recovered structures too unreliable for source-grounded explanation.

The table guards against a common pattern in neuro-symbolic research: demonstrating that a representational formalism can encode a toy structure and then assuming that the encoding explains a complex epistemic phenomenon. The relevant question is comparative. Does VSA/HDC add measurable value over well-designed symbolic graphs, neural retrievers, and graph embeddings on tasks where compositional binding and factorisation should matter?

A negative result would still be scientifically valuable. It might show that explicit graph alignment is sufficient, that open-ended codebooks defeat factorisation, or that novelty is too dependent on representation for hypervector residuals to be stable. The project should be designed so that the resulting datasets, identity theory, and lineage benchmarks remain useful even if VSA/HDC contributes little. This is the correct relation between a conceptual programme and a technical hypothesis: the hypothesis is ambitious, but the programme is not held hostage to it.

7 A Research Programme for Testing the Atlas Hypothesis

7.1 Building evidence and benchmarks without presupposing the answer

The first empirical challenge is to construct a corpus and annotation regime without turning the initial ontology into the conclusion. A dataset of "ideas" cannot be sampled in the same way as images of already named objects. Curators must decide where one contribution ends and another begins, which relations matter, and which sources count as evidence. These decisions should be treated as experimental material rather than hidden preprocessing.

A suitable corpus should begin with bounded scientific episodes. A good episode contains a tractable lineage, diverse source types, meaningful technical variation, and experts capable of reconstructing the history. Examples might include a family of optimisation methods, an experimental technique that migrated across disciplines, a causal-inference design, or a scientific concept whose terminology changed. The first objective is not representativeness of all science. It is enough variation to stress the theory of identity while preserving the possibility of close verification.

The corpus must include more than canonical papers. Reviews and textbooks reveal how fields retrospectively organise a lineage. Code and protocols reveal operational details absent from articles. Older books, proceedings, theses, patents, and archives may contain earlier attestations. Negative results and failed applications expose conditions of validity. Multilingual material is essential whenever priority or conceptual migration crosses linguistic traditions. Source selection should be documented as part of the dataset, including known gaps.

Annotation should proceed from evidence to interpretations. Curators first identify addressable attestations: passages, equations, diagrams, code fragments, or artefacts. They then propose role structures and relation claims. This order prevents the abstraction from losing its documentary anchor. Each annotation records the task and granularity at which it was produced. An expert may annotate the same passage differently for a mechanism comparison and for a historical study; both can be valid.

The protocol should collect alternative decompositions rather than force agreement too early. Inter-annotator disagreement can reveal ambiguous boundaries, ontology defects, or genuine pluralism. Conventional agreement statistics remain useful, but low agreement should not automatically be treated as annotator failure. The dataset needs a disagreement taxonomy distinguishing accidental inconsistency from different legitimate models. Adjudication should record the reasons for preferring one view in a particular benchmark.

Contrastive examples are more valuable than large quantities of easy positives. The corpus should include pairs with similar wording but different mechanisms, different wording but equivalent functions, shared mechanisms with independent histories, and apparent novelty caused by missing sources. It should also include transformations that alter only one role. These cases test whether a system has learned typed identity rather than generic semantic proximity.

Multi-resolution annotation is essential. Curators can provide a coarse operational object, a finer decomposition, and relations between them. The dataset then supports experiments on whether a model chooses an appropriate level for the question. It also enables evaluation of stability: does a claimed lineage survive refinement, or was it an artefact of coarse representation?

Historical claims require a different evidence protocol from semantic ones. An influence annotation should cite documentary or contextual support, not merely a similarity judgment. A priority annotation should describe the corpus searched and the dating basis. Independent rediscovery should remain provisional when transmission evidence is incomplete. These requirements make annotation slower, but they are necessary if the atlas is to support more than semantic search.

Existing resources provide valuable starting points. IdeaGene-Bench supplies typed objects and lineage differences across ten domains (Zhou et al. 2026). RINoBench provides expert judgments of research idea novelty (Schopf and Färber 2026). CHIMERA supplies mined recombination cases (Sternlicht and Hope 2026). Scideator and literature-grounded novelty work provide facet structures and retrieval protocols (Radensky et al. 2024; Shahid et al. 2025). The atlas research programme should reuse these resources while adding source-addressed evidence, alternative decompositions, typed uncertainty, and temporal corpus boundaries.

The resulting dataset should be regarded as a set of arguments, not only labels. A benchmark instance includes the source material, the operational representations, the relation under test, the supporting and opposing evidence, and the adjudication context. This richer form makes evaluation more demanding but reduces the risk that systems optimise against unexplained judgments.

A broad end-to-end demonstration cannot reveal which part of the atlas is scientifically responsible for an improvement. The research programme needs a family of benchmarks aligned with the conceptual distinctions developed in earlier chapters. Each benchmark should compare against strong document-level, embedding, graph, and language-model baselines.

Idea decomposition tests whether a system can propose useful role structures from sources. Evaluation should measure span grounding, role correctness, structural relations, and the ability to represent alternatives. Exact graph match is informative but too strict when several decompositions are legitimate. Pairwise expert comparison and task performance can supplement it.

Typed identity tests whether the system distinguishes paraphrase, functional equivalence, mechanistic equivalence, analogy, and lineage. The strongest cases hold topical similarity constant while changing one relation. Evaluation should score the relation, the role alignment, and the evidence cited. Calibration matters because ambiguous cases should receive distributions or alternatives rather than confident labels.

Precedent retrieval tests whether structured representations recover relevant earlier work across vocabulary and disciplinary boundaries. The benchmark must freeze the search corpus at a date and include hard negatives that are semantically close but structurally irrelevant. Recall is important because missing a decisive precedent invalidates later novelty analysis. Evidence precision is equally important because a long list of vaguely related papers does not help a researcher.

Lineage reconstruction tests inheritance and transformation. Systems receive a set of works and must identify preserved components, changes, imports, losses, and plausible transmission paths. Structural accuracy and temporal consistency can be measured automatically where annotations are strong. Historical influence should be evaluated separately from semantic lineage because the evidential standards differ.

Novelty analysis tests whether the system produces a stable and evidence-grounded profile rather than a binary score. Evaluation should ask whether the closest precedents are found, whether the role-wise differences are correct, whether uncertainty reflects corpus limitations, and whether the conclusion changes appropriately when earlier evidence is added. RINoBench shows why rationale quality and judgment accuracy must be evaluated separately (Schopf and Färber 2026).

Analogy retrieval tests relational transfer. Experts assess whether the mapping preserves causal or functional structure, whether domain incompatibilities are identified, and whether the analogy generates a testable direction. Diversity alone is not enough. A system can produce distant but scientifically empty associations. The benchmark should penalise analogies whose persuasive language exceeds their structural support.

Table 8 links each task to the scientific claim it is intended to test. This prevents benchmark proliferation from becoming disconnected from theory.

Table 8. Benchmarks should correspond to explicit scientific claims rather than to an undifferentiated demonstration.

Evaluation task	Scientific question tested
Evidence-grounded decomposition	Can operational structures be proposed reproducibly without losing their source basis or suppressing legitimate alternatives?

Evaluation task	Scientific question tested
Multi-resolution representation	Does the system choose and relate granularities appropriate to different inquiries?
Typed identity	Can it distinguish functional, mechanistic, propositional, implementation, analogical, and historical relations?
Cross-vocabulary precedent retrieval	Does explicit structure find relevant antecedents that document similarity and citations miss?
Lineage difference reconstruction	Can inheritance, transformation, import, and loss be localised to specific roles and supported by evidence?
Priority under corpus constraints	Does the system identify earlier attestations while reporting dating and coverage uncertainty?
Structured novelty analysis	Do residuals and differences align with experts and remain stable under corpus and representation changes?
Cross-domain analogy	Can relational structure transfer without uncontrolled false positives or erased boundary conditions?
Human decision support	Do researchers reason more accurately or efficiently without becoming over-reliant on system authority?

Ablation is central. The atlas should be compared with document retrieval augmented generation, dense embeddings, citation graphs, structured symbolic graphs without VSA, VSA without symbolic audit, and the full hybrid. Within VSA/HDC, experiments should vary architecture, dimension, codebook design, number of bundled roles, learned versus controlled vectors, factorisation method, and multi-view separation. Within the symbolic layer, they should vary ontology rigidity, transformation grammar, and treatment of disagreement.

The most informative failures may occur when the system is asked to generalise. A representation learned in machine learning could be tested on biology or materials science. A codebook trained on contemporary English could be tested on historical and multilingual sources. A lineage model could be tested on independent rediscoveries rather than direct citation chains. These transfers reveal whether the system has captured operational structure or only domain conventions.

Evaluation should include adversarial cases. Authors can create pseudo-novel proposals by renaming established mechanisms, combining unrelated components, or adding superficial complexity. They can also produce genuinely important local changes that appear semantically close to prior work. A useful atlas should resist both novelty inflation and novelty suppression.

7.2 Temporal backtesting, human studies, and conditions of falsification

The strongest evaluation of scientific support is temporal. Freeze the accessible corpus at time t, prevent the system from seeing later work, and ask it to analyse ideas that emerge after t. The later literature then supplies a partial record of what was independently developed, validated, abandoned, or recognised. This does not turn history into a perfect ground truth, but it is more demanding than evaluating against contemporary labels using contemporary knowledge.

Figure 6 presents the design. Competing systems receive only the frozen corpus. They perform blinded tasks such as precedent search, identity analysis, analogy, and novelty assessment. Later literature and outcomes are then revealed. Evaluation combines expert judgment, calibration, later validation, and characteristic failure analysis. The objective is not simply to predict which paper will become highly cited. It is to test whether the atlas constructs better evidence and more useful distinctions from the information historically available.

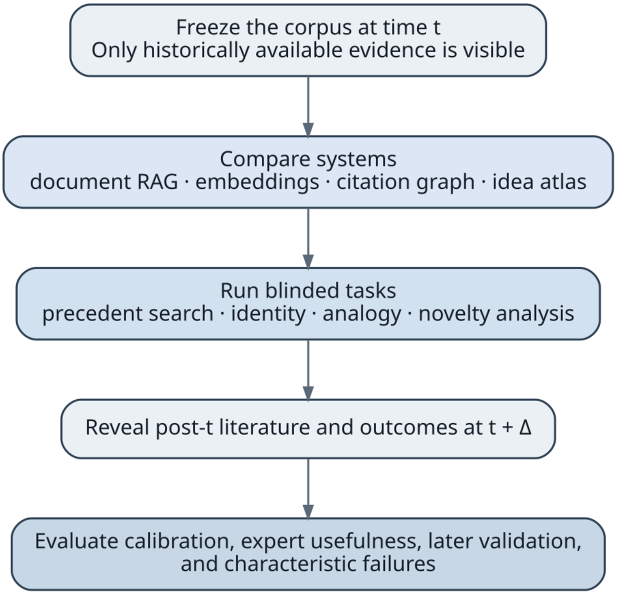

Figure 6. Temporal backtesting evaluates an atlas using only the knowledge available at a historical cutoff, then compares its analyses with later literature and expert review.

A temporal backtest can ask whether the system retrieves the antecedents later acknowledged by a field, whether it identifies an analogy that subsequently becomes productive, or whether a novelty residual corresponds to a mechanism that later receives independent support. It can also reveal false novelty caused by incomplete indexing and false analogies that later fail. Because later success is socially contingent, outcome measures should be plural: validation, replication, capability, explanatory uptake, and expert retrospective assessment.

Human studies are required because the atlas is intended to alter reasoning, not only predict labels. Researchers can be assigned literature synthesis, novelty evaluation, or hypothesis-generation tasks with different tools: ordinary search, document RAG, citation graphs, and the idea atlas. Outcomes include source coverage, correctness of distinctions, time, calibration, diversity of useful directions, and ability to detect system errors. Over-reliance must be measured explicitly by planting plausible but unsupported relations.

Expert panels should be blinded to system identity where possible. They should assess structured outputs rather than only final prose. For novelty analysis, experts can inspect retrieved precedents and role-wise differences. For analogy, they can rate mapping fidelity and testability. For lineage, historians and domain experts may need separate roles because technical equivalence and influence evidence require different expertise.

The programme must pre-register failure criteria. The identity hypothesis is weakened if multi-view structured representations do not outperform strong embeddings on controlled contrastive cases. The lineage hypothesis is weakened if evidence-grounded graphs do not improve reconstruction beyond citation and text baselines. The novelty hypothesis is weakened if residuals are unstable under reasonable changes of corpus and representation or mainly predict unusual wording. The VSA hypothesis is weakened if symbolic graphs supply the same gains with less noise and greater auditability.

A stronger failure would be institutional. Even accurate outputs may not improve research if they are too complex, slow, or difficult to challenge. Researchers may ignore uncertainty and treat ranked precedents as definitive. Curators may be unable to maintain the ontology. Coverage bias may dominate the results. These are not secondary deployment issues; they test whether the epistemic model can function in real scientific communities.

Negative results should be publishable and reusable. If operational decomposition proves reliable but VSA factorisation fails, the dataset and identity theory remain contributions. If the atlas improves review but not ideation, that finding narrows the impact claim. If document-level systems match the hybrid on all tasks, the proposed epistemic layer may not justify its complexity. A rigorous programme is one in which each outcome teaches us something about the nature and representation of ideas.

The research agenda is therefore ambitious but falsifiable. It does not ask whether a system can produce an impressive genealogy for selected examples. It asks whether explicit idea structure yields reproducible gains, whether the gains survive domain and time shifts, whether uncertainty behaves appropriately, and whether humans reason better with the instrument. Only after these conditions are met would claims about accelerating science be warranted.

8 Where a Working Atlas Would Matter

8.1 Scientific memory, synthesis, and attribution

The first practical impact of an idea atlas would not be automated discovery. It would be a change in scientific memory. Today, literature synthesis is forced to move repeatedly between documents and implicit conceptual models held by readers. Reviewers and authors reconstruct which mechanisms recur, which assumptions changed, and how results relate, but those reconstructions are rarely preserved in reusable form. A working atlas would make the intermediate reasoning available for inspection and cumulative improvement.

Systematic reviews could represent not only interventions and outcomes but the operational ideas connecting them. Studies that appear heterogeneous at the level of terminology might share a mechanism; studies grouped under one method name might in fact implement different causal structures. An idea-centred representation could help reviewers decide when evidence should be combined, when subgroup analysis is required, and where apparent disagreement results from different assumptions. The benefit would depend on domain-specific validation and should not replace established review protocols.

Narrative reviews and textbooks would gain a different capability. Instead of presenting a lineage as a static story, they could expose alternative decompositions and disputed continuities. Readers could move from a general principle to its historical formulations, technical refinements, applications, and known failures. The map would show not only who cited whom but what was inherited or changed. This could make review writing more accountable and reduce the repeated reconstruction of the same intellectual history.

Laboratory memory is another important setting. Research groups accumulate tacit knowledge about failed parameter regimes, abandoned hypotheses, local modifications, and reasons for design choices. Much of it disappears when members leave or survives only in notebooks and conversations. An atlas linked to protocols, code, and results could represent why a method was attempted, which earlier idea it instantiated, and which boundary condition caused failure. Capturing negative results would help prevent unproductive repetition and improve the interpretation of later successes.

Attribution and priority research could become more nuanced. The system could search across languages and source types for earlier attestations, separate first formulation from first validation, and distinguish structural similarity from evidence of influence. Historians would still adjudicate the conclusions, but the search space would become more tractable. A conservative and transparent atlas might correct canonical narratives that currently reflect visibility more than chronology.

Peer review could use the atlas as an evidence instrument. A reviewer evaluating a novelty claim could examine the closest structured precedents and the role-wise differences. This could reduce superficial novelty based on terminology and help recognise important local changes that ordinary similarity underestimates. The risk is obvious: institutions may turn a support tool into an automated gatekeeper. The atlas should therefore present evidence and alternative interpretations, not a pass–fail novelty score.

Grant evaluation and research portfolio analysis present a similar tension. An idea map could identify duplicated mechanisms, neglected combinations, and clusters that share hidden assumptions. Funders might use it to diversify portfolios or find high-information experiments. They might also use it to penalise unconventional proposals that lack recognised antecedents. The technology's impact depends on whether it expands the space of inquiry or narrows it to what the ontology already understands.

Table 9 pairs major academic settings with the transformation a successful atlas could enable. The second column also states the main condition that prevents the application from becoming a simple automation story.

Table 9. Academic impact would arise from better conceptual memory and comparison, provided the atlas remains evidence-centred and contestable.

Academic setting	Potential transformation and necessary condition
Systematic and evidence reviews	Align studies by mechanism, assumptions, and evidential role; domain validation must determine when structural similarity justifies synthesis.
Narrative reviews and textbooks	Preserve alternative lineages and multi-resolution explanations; editorial interpretation must remain visible and contestable.
Laboratory and organisational memory	Connect hypotheses, design choices, failures, code, and protocols to reusable ideas; tacit and negative knowledge must be captured without false precision.
Peer review	Provide structured precedents and localised differences; the system must support judgment rather than issue authoritative novelty verdicts.
Grant and portfolio analysis	Reveal conceptual duplication, neglected mechanisms, and strategic diversity; uncertainty and ontology bias must be explicit.
Historical attribution	Search for earlier attestations and separate priority from influence; corpus coverage and dating evidence must accompany every claim.

Academic setting	Potential transformation and necessary condition
Research integrity analysis	Distinguish textual copying, conceptual reuse, common method, and independent rediscovery; similarity alone cannot establish misconduct.
Scientific education	Teach concepts through transformations, applications, and failure boundaries; multiple traditions must be represented rather than reduced to a single canon.

The table shows that the practical applications share one underlying function: preservation of intermediate epistemic structure. They do not require the system to generate new ideas. Even if discovery support proves limited, a reliable instrument for synthesis and attribution could have substantial value. This is an important asymmetry in the impact case. The atlas can be useful before it becomes creative.

8.2 Discovery, cross-domain transfer, and institutional consequences

The most ambitious application is to support the generation of scientifically useful ideas. Existing language-model systems already produce proposals, and studies show that researchers can find some of them novel or inspiring (Si, Yang, and Hashimoto 2024; Radensky et al. 2024). The difficult question is whether structured memory improves the quality of the search rather than merely the persuasiveness of the output.

An idea atlas could support discovery in several ways. It could reveal unresolved residuals in a lineage, such as a persistent assumption that no work has removed. It could identify mechanisms repeatedly successful under one class of constraints and search for domains with analogous structure. It could locate components that have never been combined despite compatible roles. It could also identify where evidence lags behind theory, suggesting experiments with high information value.

Cross-domain transfer is especially promising because disciplines often use incompatible vocabularies for similar structures. Analogy research suggests that productive transfer depends on relational mapping rather than surface resemblance (Gentner 1983). An atlas with mechanistic and functional views could search for distant cases that share control structure, feedback, invariance, resource constraints, or measurement problems. It should then expose the mismatches that could invalidate transfer. The value lies as much in explaining why an analogy may fail as in generating it.

Recombination systems such as CHIMERA, Scideator, and Graph2Idea demonstrate that structured facets and relational contexts can guide ideation (Sternlicht and Hope 2026; Radensky et al. 2024; Li, Tu, and Han 2026). The atlas would extend this work by adding lineage, identity types, negative evidence, and transformation grammars. Instead of asking only which components can be combined, it could ask what form of change has historically overcome a similar limitation and whether the enabling assumptions exist in the target domain.

The system could also improve abductive reasoning. Given an anomalous result, it might retrieve mechanisms that produce an analogous pattern elsewhere, align their assumptions, and suggest discriminating tests. Because the atlas records evidence and failure conditions, hypotheses can be ranked by explanatory fit and by the cost of experiments that separate them. Such support would be closer to scientific reasoning than unconstrained text generation.

For artificial intelligence, the atlas could function as an external memory and a training resource. Document corpora supply abundant language but sparse supervision for conceptual relations. The atlas would provide pairs and lineages labelled with paraphrase, functional identity, mechanistic change, independent rediscovery, and evidence type. Models could be trained contrastively to preserve these distinctions and evaluated on whether they recover the correct relation rather than merely produce fluent explanations.

The provenance layer could improve retrieval-augmented generation. Instead of retrieving entire documents, an AI system could retrieve a structured idea object together with its supporting attestations, competing interpretations, and known limitations. This would reduce context redundancy and make citations more local. The language model would still generate explanations, but the claims would be constrained by inspectable evidence.

A scientific agent could use the atlas to maintain a research state. It would record which components of a hypothesis are inherited, which are speculative, which experiments bear on them, and how new evidence changes the structure. This is preferable to a conversational memory that stores only prose summaries. The agent's reasoning could be evaluated against the explicit transformations it proposes.

Training on an atlas also presents risks. The representation may harden existing categories and make models less able to recognise ideas outside the ontology. Widely accepted but mistaken lineages could become high-quality supervision. If the atlas overrepresents successful work, agents may learn to imitate recognised trajectories and avoid high-risk exploration. Training should therefore include disagreements, rejected relations, negative results, and examples where the system's initial identity model was revised.

The impact on scientific acceleration should be measured in experiments, not inferred from the sophistication of the representation. Relevant outcomes include time to find decisive precedents, quality of hypotheses, value of selected experiments, replication success, and the ability to transfer mechanisms across fields. Citation impact is too delayed and socially confounded to serve as the only measure. A system that helps researchers abandon a bad idea earlier may accelerate science without producing a highly cited paper.

A technology that maps identity, novelty, and lineage would not be neutral infrastructure. It would influence how credit is assigned, which proposals appear original, and which traditions are visible. The closer it comes to working, the greater the governance burden. The central risk is not only technical error but the conversion of contestable interpretations into institutional facts.

Credit is particularly sensitive. A priority map can restore overlooked contributions, but it can also produce false precision. An early vague resemblance may be promoted over the first clear formulation; a dominant-language source may appear earliest because other corpora are missing; a collective tradition may be reduced to an individual node. The atlas should therefore represent several forms of precedence and avoid translating them automatically into authorship or ownership.

Research evaluation could be distorted by metricisation. Once novelty profiles exist, institutions may seek a single score for hiring, funding, or publication. This would invite optimisation against the measure. Authors could maximise unusual combinations, manipulate the representation level, or strategically frame familiar work as a new mechanism. The system should make such compression difficult by refusing a universal score and by exposing the underlying comparisons.

Openness creates a second tension. An open atlas supports correction, replication, and broad access. Yet some scientific ideas involve security, privacy, dual use, or proprietary knowledge. The provenance system must represent access restrictions without treating unavailable evidence as nonexistent. Governance may require tiered access, embargoes, and policies for sensitive lineages. These measures can reduce transparency, so their effects should be recorded.

The atlas could alter intellectual property practice. Prior-art search might become more semantically powerful, and firms could map transferable mechanisms across patents and scientific literature. This may reduce weak claims based on terminology. It may also favour organisations with the resources to curate proprietary corpora and challenge public priority records. Public research infrastructure would be important to prevent conceptual search from becoming exclusively private.

Education could benefit from maps that connect principles to their transformations and applications. Students often encounter results as isolated named techniques. An idea atlas could show why a method was introduced, which limitation it addressed, and how it changed. The risk is canon formation: the map may present one dominant lineage as inevitable. Educational use should include alternative traditions, failed paths, and disputes about identity.

At a societal level, a working atlas could improve collective problem solving by exposing structural analogies between scientific, engineering, and policy domains. However, transfer into social systems is especially hazardous because agents react to interventions and concepts carry normative meanings. Mechanistic analogies from physics or computation can become misleading metaphors when institutional context is ignored. The system must represent domain incompatibilities, not only similarities.

The long-term potential is substantial. Scientific literature could become a substrate for explicit reasoning about transformation rather than a warehouse of documents. AI systems could be trained on conceptual inheritance and evidential change. Researchers could inspect how an idea travelled, where it failed, and which unresolved variants remain. But this potential depends on cultural practices as much as algorithms. Communities must value correction, preserve negative evidence, and tolerate multiple representations.

A successful atlas would therefore have an unusual form of impact. It would not simply automate a task. It would create a new public object: the contestable, evidence-grounded representation of an idea and its transformations. Such objects could become part of scholarly communication in the way datasets and code have become part of many fields. Their value would lie in allowing scientific communities to reason collectively about continuity and change while retaining the sources and disagreements that make those judgments credible.

Conclusion: A Microscope for Ideas

The vision examined in this book begins with a straightforward dissatisfaction. Science stores and retrieves documents, while many of its most important questions concern structures that cross document boundaries. Researchers want to know whether two contributions contain the same mechanism, how a method changed, whether an apparent novelty has an earlier precedent, and where a useful idea might transfer. These questions are currently answered through expert reading, informal conceptual models, and increasingly through language models whose fluency can exceed the reliability of their judgments.

An atlas of operational ideas would insert a new epistemic layer. Its units would not be timeless atoms discovered once and for all. They would be task-relative models grounded in addressable evidence. They would record goals, mechanisms, assumptions, contexts, outcomes, evidence, and provenance at several levels of resolution. Their relations would be typed: paraphrase, functional equivalence, mechanistic correspondence, analogy, specialisation, recombination, influence, and independent rediscovery would no longer be compressed into generic similarity.

This pluralism is not a retreat from formalisation. It is the condition for a more adequate formalisation. Identity can be represented as a profile of preserved and transformed roles. Lineage can be reconstructed as a defeasible alignment supported by documentary and temporal evidence. Priority can be stated as the earliest known attestation within an explicit corpus rather than as an absolute first thought. Novelty can be analysed relative to a dated knowledge state and a declared transformation grammar. Usefulness can remain separate from originality and be assessed through explanatory, predictive, experimental, operational, and generative consequences.

The neuro-symbolic architecture follows from these commitments. Neural models provide open-ended semantic reach. Symbolic structures preserve roles, relations, constraints, and arguments. Probabilistic and temporal models represent uncertainty and incomplete observation. Human experts contribute interpretation and challenge. The architecture is not valuable because it combines fashionable components. It is valuable only if the components correct one another and if a conclusion remains traceable after any one model is replaced.

VSA/HDC offers an especially interesting but unproven hypothesis. Role–filler binding and superposition could support multi-view representations that preserve structure while allowing approximate retrieval. Factorisation could search for antecedent components and recurrent transformations. Persistent role-specific reconstruction residuals may correlate with substantive novelty. Yet the same residual can arise from missing literature, codebook failure, wrong granularity, extraction error, or incoherence.

The VSA layer must therefore remain an experimental instrument inside a source-audited system.

The research programme is feasible because it can be broken into falsifiable questions. Can experts construct operational decompositions with useful agreement while preserving alternatives? Do typed representations outperform strong embeddings on contrastive identity tasks? Does structure improve retrieval of cross-vocabulary precedents? Can lineages be reconstructed without confusing resemblance with influence? Are novelty residuals stable under corpus and representation changes? Does VSA/HDC add value beyond symbolic graphs and neural retrieval? Do researchers make better decisions with the atlas, and can they detect when it is wrong?

A negative answer to some of these questions would not invalidate the entire programme. The identity theory, datasets, provenance structures, and lineage benchmarks could remain useful even if novelty estimation proves unstable or VSA factorisation fails. This modularity is a scientific strength. It prevents the project from depending on one grand technical promise.

The practical impact of a working atlas would begin with memory and synthesis. Reviews, laboratories, textbooks, peer evaluation, and historical attribution all reconstruct idea structures that are currently lost after use. Preserving those structures could make scholarship more cumulative. Discovery support and AI training are more ambitious extensions. They become credible only when the system demonstrates that it can find decisive precedents, support valid analogies, and improve human reasoning under blinded and temporal evaluation.

The technology would also create power. Whoever controls the map may influence what counts as the same idea, what appears new, and who receives credit. The atlas must therefore be open to alternative representations, explicit about coverage, conservative in high-stakes claims, and designed for challenge and revision. It should resist becoming a metric of originality or a central authority over scientific history.

The best metaphor is a microscope rather than an oracle. A microscope does not decide what a specimen means. It makes structure visible under controlled conditions, introduces artefacts that must be understood, and improves through calibration and comparison. An atlas of operational ideas could do the same for scientific thought. It would make inheritance, transformation, and evidential difference more inspectable, while reminding its users that every view depends on a lens.

The long-term scientific opportunity is not to prove that all ideas have a universal essence. It is to build representations precise enough to support reasoning and humble enough to remain corrigible. If that balance can be achieved, the atlas could become a new kind of knowledge infrastructure: one that helps humans and machines navigate not only what has been written, but how the actionable structures of knowledge persist, change, combine, and occasionally become genuinely new.

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